

Commutative Algebra 1

Problem 1 — Chinese remainder theorem for ideals (kernel/surjectivity/product vs intersection)

Exact statement. We say two ideals I, J of a ring A are *coprime* if $I + J = (1) = A$. Given ideals $I_1, \dots, I_n$ of A , there is a natural map

$$\phi : A \longrightarrow \prod_{i=1}^n A/I_i, \quad \phi(a) = (a + I_1, \dots, a + I_n).$$

Show the following:

(a) If I_i, I_j are coprime whenever $i \neq j$, then

$$\bigcap_{i=1}^n I_i = \prod_{i=1}^n I_i.$$

[Note: the ideal product IJ is the ideal generated by elements ab with $a \in I, b \in J$.]

(b) ϕ is surjective $\iff I_i, I_j$ are coprime whenever $i \neq j$.

(c) ϕ is injective $\iff \bigcap_{i=1}^n I_i = 0$.

Solution. (c) We compute the kernel:

$$\phi(a) = 0 \iff (a + I_i = 0 \text{ for all } i) \iff (a \in I_i \text{ for all } i) \iff a \in \bigcap_{i=1}^n I_i.$$

Hence $\ker(\phi) = \bigcap_{i=1}^n I_i$, so ϕ is injective iff $\bigcap_{i=1}^n I_i = 0$.

(a) First recall the 2-ideal lemma: if $I + J = A$, then $I \cap J = IJ$. Indeed, $IJ \subseteq I \cap J$ always. Conversely, pick $x \in I, y \in J$ with $x + y = 1$. If $z \in I \cap J$, then

$$z = z(x + y) = zx + zy.$$

Here $zx \in JI = IJ$ and $zy \in IJ$, so $z \in IJ$. Thus $I \cap J = IJ$.

Now assume $I_i + I_j = A$ for all $i \neq j$. We prove by induction that $\bigcap_{i=1}^n I_i = \prod_{i=1}^n I_i$. For the induction step, set $P_k = \prod_{i=1}^k I_i$. From $I_i + I_{k+1} = A$ choose $a_i \in I_i, b_i \in I_{k+1}$ with $a_i + b_i = 1$ for $i = 1, \dots, k$. Then

$$\prod_{i=1}^k (a_i + b_i) = 1.$$

Expanding, the term $\prod_{i=1}^k a_i$ lies in P_k , and every other term contains some $b_i \in I_{k+1}$, hence lies in I_{k+1} . So $1 = u + v$ with $u \in P_k$, $v \in I_{k+1}$, i.e. $P_k + I_{k+1} = A$. By the 2-ideal lemma,

$$\left(\bigcap_{i=1}^k I_i\right) \cap I_{k+1} = P_k \cap I_{k+1} = P_k I_{k+1} = \prod_{i=1}^{k+1} I_i,$$

using the induction hypothesis $\bigcap_{i=1}^k I_i = P_k$. This proves (a).

(b) ($\Rightarrow$) If ϕ is surjective, then for any $i \neq j$ the composition of ϕ with the projection $\prod A/I_\ell \rightarrow A/I_i \times A/I_j$ is surjective. The two-ideal CRT says $A \rightarrow A/I_i \times A/I_j$ is surjective iff $I_i + I_j = A$. Hence I_i and I_j are coprime.

($\Leftarrow$) Assume $I_i + I_j = A$ for all $i \neq j$. Fix i and let $J_i = \bigcap_{j \neq i} I_j$. As in (a) (using the same product expansion trick), $I_i + J_i = A$. Choose $e_i \in J_i$, $f_i \in I_i$ with $e_i + f_i = 1$. Then $e_i \equiv 1 \pmod{I_i}$ and $e_i \equiv 0 \pmod{I_j}$ for $j \neq i$.

Given $(\bar{a}_1, \dots, \bar{a}_n) \in \prod_{i=1}^n A/I_i$, pick representatives $a_i \in A$ and define

$$a = \sum_{i=1}^n a_i e_i.$$

Modulo I_k , every term with $i \neq k$ vanishes since $e_i \in I_k$, and $e_k \equiv 1 \pmod{I_k}$, so $a \equiv a_k \pmod{I_k}$. Hence $\phi(a) = (\bar{a}_1, \dots, \bar{a}_n)$, proving surjectivity.

Problem 2 — Tensoring with A/I kills IM

Exact statement. Let A be a ring, $I \subseteq A$ an ideal, and M an A -module. Show that

$$(A/I) \otimes_A M \cong M/IM.$$

Solution. Define $\Psi : (A/I) \otimes_A M \rightarrow M/IM$ on pure tensors by

$$\Psi(\bar{a} \otimes m) = am + IM.$$

This is well-defined: if $a - a' \in I$, then $(a - a')m \in IM$, so $am + IM = a'm + IM$. It is A -balanced, hence induces an A -module homomorphism. It is surjective since $\Psi(\bar{1} \otimes m) = m + IM$.

Define $\Theta : M \rightarrow (A/I) \otimes_A M$ by $\Theta(m) = \bar{1} \otimes m$. If $m \in IM$, say $m = \sum_k i_k m_k$ with $i_k \in I$, then

$$\Theta(m) = \sum_k \bar{1} \otimes i_k m_k = \sum_k \bar{i}_k \otimes m_k = 0,$$

so $IM \subseteq \ker \Theta$. Thus Θ factors to

$$\bar{\Theta} : M/IM \rightarrow (A/I) \otimes_A M, \quad m + IM \mapsto \bar{1} \otimes m.$$

Now

$$\Psi(\bar{\Theta}(m + IM)) = \Psi(\bar{1} \otimes m) = m + IM,$$

so $\Psi \circ \bar{\Theta} = \text{id}$. Also for pure tensors,

$$\bar{\Theta}(\Psi(\bar{a} \otimes m)) = \bar{\Theta}(am + IM) = \bar{1} \otimes (am) = \bar{a} \otimes m,$$

so $\bar{\Theta} \circ \Psi = \text{id}$. Hence Ψ is an isomorphism and $(A/I) \otimes_A M \cong M/IM$.

Theory — Ideals, quotients, CRT, and tensoring

Ideals and quotients. An ideal $I \subseteq A$ is an additive subgroup with $aI \subseteq I$ for all $a \in A$. The quotient ring A/I consists of cosets $a + I$; elements of I become 0. Universal property: any ring map $f : A \rightarrow B$ with $I \subseteq \ker f$ factors uniquely through A/I .

Products and intersections. For ideals I, J , the product IJ is the ideal generated by finite sums $\sum_k a_k b_k$ with $a_k \in I, b_k \in J$. Always $IJ \subseteq I \cap J$. If $I + J = A$ (coprime), then $I \cap J = IJ$; indeed, pick $x \in I, y \in J$ with $x + y = 1$, and for $z \in I \cap J$ we have $z = z(x + y) = zx + zy \in IJ$.

Chinese remainder theorem (ideals). Given $I_1, \dots, I_n$, define

$$\phi : A \rightarrow \prod_{i=1}^n A/I_i, \quad \phi(a) = (a + I_1, \dots, a + I_n).$$

Then $\ker \phi = \bigcap_{i=1}^n I_i$. If $I_i + I_j = A$ for all $i \neq j$, then ϕ is surjective and

$$A / \bigcap_{i=1}^n I_i \cong \prod_{i=1}^n A/I_i, \quad \text{and in fact } \bigcap_{i=1}^n I_i = \prod_{i=1}^n I_i.$$

A constructive proof uses elements e_i with $e_i \equiv 1 \pmod{I_i}$ and $e_i \equiv 0 \pmod{I_j}$ for $j \neq i$, so that solutions are $a = \sum_i a_i e_i$.

Tensor products and reduction mod I . For an A -module M , the tensor product $(A/I) \otimes_A M$ is the universal recipient of A -balanced maps $(A/I) \times M \rightarrow N$. Since $I = 0$ in A/I , tensoring forces $i \cdot m$ to become 0 for all $i \in I$, so the submodule IM is killed. Precisely,

$$(A/I) \otimes_A M \cong M/IM,$$

via $\bar{a} \otimes m \mapsto am + IM$, with inverse $m + IM \mapsto \bar{1} \otimes m$.

Outer products vs tensors (linear algebra intuition). Over a field, choosing bases identifies $v \otimes w \in V \otimes W$ with the rank-one matrix vw^T ; thus the usual “outer product” is a coordinate realization of a tensor.

Examples — Problem 1 (CRT for ideals)

Example 1 ($\mathbb{Z}$: 3 and 4). Let $A = \mathbb{Z}$, $I_1 = (3)$, $I_2 = (4)$. Then $I_1 + I_2 = \mathbb{Z}$. The map $\phi(a) = (a \bmod 3, a \bmod 4)$ is surjective and $\ker \phi = (3) \cap (4) = (12)$, hence $\mathbb{Z}/12 \cong \mathbb{Z}/3 \times \mathbb{Z}/4$. To solve $x \equiv 2 \pmod{3}$, $x \equiv 1 \pmod{4}$, take $e_1 = 4$ and $e_2 = 9$, then $x = 2e_1 + 1e_2 = 17 \equiv 5 \pmod{12}$.

Example 2 (not coprime: 4 and 6). Let $A = \mathbb{Z}$, $I_1 = (4)$, $I_2 = (6)$. Then $I_1 + I_2 = (2) \neq \mathbb{Z}$, so $\phi : \mathbb{Z} \rightarrow \mathbb{Z}/4 \times \mathbb{Z}/6$ is not surjective, e.g. $(1 \bmod 4, 0 \bmod 6)$ has no preimage. Also $(4) \cap (6) = (12) \neq (24) = (4)(6)$.

Example 3 ($k[x]$: (x) and $(x-1)$). Let $A = k[x]$, $I_1 = (x)$, $I_2 = (x-1)$. Since $(x) + (x-1) = A$, CRT gives $k[x]/(x(x-1)) \cong k \times k$ via $f \mapsto (f(0), f(1))$. Given $a, b \in k$, $f(x) = a(1-x) + bx$ satisfies $f(0) = a$, $f(1) = b$.

Example 4 (selectors in $\mathbb{Z}/12$). In $A = \mathbb{Z}/12$, ideals (4) and (3) satisfy $(4) + (3) = A$. Elements $e_1 = 9$ and $e_2 = 4$ obey $e_1 \equiv 1 \pmod{(4)}$, $e_1 \equiv 0 \pmod{(3)}$ and $e_2 \equiv 0 \pmod{(4)}$, $e_2 \equiv 1 \pmod{(3)}$, hence $e_1^2 = e_1$, $e_2^2 = e_2$, $e_1 e_2 = 0$, $e_1 + e_2 = 1$.

Examples — Problem 2 $(A/I) \otimes_A M \cong M/IM$

Example 1 ($\mathbb{Z}$: free module). Let $A = \mathbb{Z}$, $I = (m)$, $M = \mathbb{Z}^r$. Then $(\mathbb{Z}/m) \otimes_{\mathbb{Z}} \mathbb{Z}^r \cong \mathbb{Z}^r/m\mathbb{Z}^r \cong (\mathbb{Z}/m)^r$.

Example 2 ($\mathbb{Z}$: torsion). Let $A = \mathbb{Z}$, $I = (m)$, $M = \mathbb{Z}/n$. Then $(\mathbb{Z}/m) \otimes_{\mathbb{Z}} (\mathbb{Z}/n) \cong \mathbb{Z}/\gcd(m, n)$.

Example 3 ($k[x]$: evaluate at $x = 0$). Let $A = k[x]$, $I = (x)$. Then $A/I \cong k$ and for any A -module M , $k \otimes_{k[x]} M \cong M/xM$. In particular $k \otimes_{k[x]} k[x] \cong k$.

Example 4 ($k[x]$: $M = A/(x^2)$). Let $A = k[x]$, $I = (x)$, $M = A/(x^2)$. Then $(A/I) \otimes_A M \cong M/xM \cong k$.

Problem 4 — Tensor products of k -algebras

Exact statement. Calculate the following tensor products of rings, where k is a field:

- (a) $k \otimes_{k[x,y]} k[u, v]$, where $k[x, y] \rightarrow k$ is $x, y \mapsto 0$, and $k[x, y] \rightarrow k[u, v]$ is $x \mapsto u, y \mapsto uv$, both the identity on k .
- (b) $k[v] \otimes_{k[x]} k[v]$, where both maps $k[x] \rightarrow k[v]$ send $x \mapsto v^2$ and are the identity on k .

Solution. (a) Since $k \cong k[x, y]/(x, y)$, we use base change:

$$k \otimes_{k[x,y]} k[u, v] \cong (k[x, y]/(x, y)) \otimes_{k[x,y]} k[u, v] \cong k[u, v]/(x, y) k[u, v].$$

Here (x, y) acts on $k[u, v]$ by $x \mapsto u, y \mapsto uv$, so

$$(x, y) k[u, v] = (u, uv) = (u),$$

hence

$$k \otimes_{k[x,y]} k[u, v] \cong k[u, v]/(u) \cong k[v].$$

(b) Distinguish the two copies as $k[v]$ and $k[w]$. Both are $k[x]$ -algebras via $x \mapsto v^2$ and $x \mapsto w^2$. Therefore

$$k[v] \otimes_{k[x]} k[w] \cong k[v, w]/(v^2 - w^2).$$

If $\text{char}(k) \neq 2$, then $v^2 - w^2 = (v - w)(v + w)$ and $(v - w) + (v + w) = (1)$, so CRT gives

$$k[v, w]/(v^2 - w^2) \cong k[v, w]/(v - w) \times k[v, w]/(v + w) \cong k[v] \times k[v].$$

If $\text{char}(k) = 2$, then $v^2 - w^2 = (v - w)^2$, so $k[v, w]/(v^2 - w^2) = k[v, w]/(v - w)^2$ is nonreduced.

Problem 5 — $A[x]$ is flat over A

Exact statement. Let $A[x]$ denote the polynomial ring in one variable over a ring A . Show that $A[x]$ is a flat A -module.

Solution. As an A -module,

$$A[x] \cong \bigoplus_{n \geq 0} A \cdot x^n,$$

so $A[x]$ is free with basis $\{1, x, x^2, \dots\}$. Every free module is flat. Hence $A[x]$ is flat over A .

Problem 6 — Properties of localization

Exact statement.

- (a) Let $S \subseteq A$ be multiplicatively closed, and M a finitely generated A -module. Show $S^{-1}M = 0$ iff there exists $s \in S$ such that $sM = 0$.
- (b) Let $S, T \subseteq A$ be multiplicatively closed, and let U be the image of T in $S^{-1}A$. Show $(ST)^{-1}A \cong U^{-1}(S^{-1}A)$, where $ST = \{st \mid s \in S, t \in T\}$.
- (c) Let $f : A \rightarrow B$ be a ring homomorphism, $S \subseteq A$ multiplicatively closed, and $T = f(S)$. Show $S^{-1}B \cong T^{-1}B$ as $S^{-1}A$ -modules.

Solution. (a) If $sM = 0$ for some $s \in S$, then for any $m \in M$,

$$\frac{m}{1} = \frac{sm}{s} = 0$$

in $S^{-1}M$, so $S^{-1}M = 0$. Conversely, suppose $S^{-1}M = 0$ and M is generated by $m_1, \dots, m_r$. Then $\frac{m_i}{1} = 0$ implies $\exists s_i \in S$ with $s_i m_i = 0$. Let $s = s_1 \cdots s_r \in S$. Then $sm_i = 0$ for all i , hence $sM = 0$.

(b) Define

$$\Phi : (ST)^{-1}A \rightarrow U^{-1}(S^{-1}A), \quad \Phi\left(\frac{a}{st}\right) = \frac{a/s}{t},$$

and

$$\Psi : U^{-1}(S^{-1}A) \rightarrow (ST)^{-1}A, \quad \Psi\left(\frac{a/s}{t}\right) = \frac{a}{st}.$$

These maps are well-defined ring homomorphisms and satisfy $\Psi \circ \Phi = \text{id}$ and $\Phi \circ \Psi = \text{id}$. Hence $(ST)^{-1}A \cong U^{-1}(S^{-1}A)$.

(c) Define

$$\alpha : S^{-1}B \rightarrow T^{-1}B, \quad \alpha\left(\frac{b}{s}\right) = \frac{b}{f(s)}.$$

This is well-defined and $S^{-1}A$ -linear. Conversely, the map $B \rightarrow S^{-1}B$ sends every $f(s) \in T$ to a unit, so it extends uniquely to a map $\beta : T^{-1}B \rightarrow S^{-1}B$ by the universal property of localization:

$$\beta\left(\frac{b}{f(s)}\right) = \frac{b}{s}.$$

Then α and β are inverse isomorphisms, so $S^{-1}B \cong T^{-1}B$ as $S^{-1}A$ -modules.

Problem 7 — Localizing $k[x, y, z]/(xy - z^2)$ at x

Exact statement. Prove that

$$\left(\frac{k[x, y, z]}{(xy - z^2)}\right)_x \cong k[x, z]_x,$$

where $(\cdot)_x$ denotes localization at $\{1, x, x^2, \dots\}$.

Solution. Let $R = k[x, y, z]/(xy - z^2)$. In R_x , the element x is invertible, so the relation $xy = z^2$ gives

$$y = \frac{z^2}{x} \quad \text{in } R_x.$$

Define $\varphi : k[x, z]_x \rightarrow R_x$ by $\varphi(x) = \bar{x}$, $\varphi(z) = \bar{z}$. Then φ is surjective because $\bar{y} = \bar{z}^2/\bar{x} \in \text{im}(\varphi)$.

Define $\psi : R \rightarrow k[x, z]_x$ by $\psi(x) = x$, $\psi(z) = z$, $\psi(y) = z^2/x$. Then $\psi(xy - z^2) = x(z^2/x) - z^2 = 0$, so ψ factors through R . Since $\psi(x)$ is invertible in $k[x, z]_x$, it extends to $\psi_x : R_x \rightarrow k[x, z]_x$. One checks $\psi_x \circ \varphi = \text{id}$ and $\varphi \circ \psi_x = \text{id}$, hence $R_x \cong k[x, z]_x$.

Theory — Problems 4–7 (tensor products, flatness, localization)

Theory. Tensor products of rings. Given ring maps $A \rightarrow B$ and $A \rightarrow C$, the tensor product $B \otimes_A C$ is the *pushout* in the category of commutative A -algebras: it is universal among A -algebras receiving maps from B and C that agree on A . Geometrically,

$$\text{Spec}(B \otimes_A C) \cong \text{Spec}(B) \times_{\text{Spec}(A)} \text{Spec}(C).$$

A key computation rule (base change by a quotient) is:

$$(A/I) \otimes_A B \cong B/IB,$$

where IB is the ideal generated by the image of I in B . Thus tensor products often amount to imposing relations inside the target ring. In characteristic issues, $k[v, w]/(v^2 - w^2)$ is reduced and splits when $\text{char}(k) \neq 2$ (CRT), but becomes nonreduced when $\text{char}(k) = 2$.

Theory. Flatness and $A[x]$. An A -module N is *flat* if tensoring with N preserves exactness (equivalently, preserves injections). Free modules are flat, and direct sums of flat modules are flat. Since

$$A[x] \cong \bigoplus_{n \geq 0} A \cdot x^n$$

as an A -module, it is free with basis $\{1, x, x^2, \dots\}$, hence flat.

Theory. Localization. For a multiplicatively closed set $S \subseteq A$, the localization $S^{-1}A$ is characterized by the universal property that elements of S become units. For modules,

$$S^{-1}M \cong S^{-1}A \otimes_A M,$$

and localization is exact because $S^{-1}A$ is flat. Clearing denominators gives

$$\frac{m}{s} = 0 \iff \exists t \in S \text{ such that } tm = 0 \text{ in } M.$$

Finite generation is essential in statements like: $S^{-1}M = 0 \Rightarrow \exists s \in S$ with $sM = 0$, because one needs a single denominator that kills a finite generating set. Iterated localization satisfies $(ST)^{-1}A \cong U^{-1}(S^{-1}A)$ (same universal property), and for $f : A \rightarrow B$ one has $S^{-1}B \cong (f(S))^{-1}B$.

Theory. Localizing $k[x, y, z]/(xy - z^2)$ at x . Let $R = k[x, y, z]/(xy - z^2)$. Localizing at x corresponds to restricting to $D(x)$ where x is invertible. Then the relation $xy = z^2$ gives $y = z^2/x$, so y becomes redundant and

$$R_x \cong k[x, z]_x.$$

Examples pack — tensors, flatness, localization

A. Module tensor products

(A1) $\mathbb{Z}^r \otimes_{\mathbb{Z}} M \cong M^r$. Example: $\mathbb{Z}^3 \otimes \mathbb{Z}/5 \cong (\mathbb{Z}/5)^3$.

(A2) $(\mathbb{Z}/m) \otimes_{\mathbb{Z}} M \cong M/mM$. Example: if $M = \mathbb{Z} \oplus \mathbb{Z}/6$ and $m = 4$, then $(\mathbb{Z}/4) \otimes M \cong (\mathbb{Z}/4) \oplus (\mathbb{Z}/2)$.

(A3) $(\mathbb{Z}/m) \otimes (\mathbb{Z}/n) \cong \mathbb{Z}/\gcd(m, n)$. Example: $(\mathbb{Z}/12) \otimes (\mathbb{Z}/8) \cong \mathbb{Z}/4$.

(A4) Tensor can kill injections: $0 \rightarrow \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z}$ is injective, but after tensoring with $\mathbb{Z}/2$ becomes $0 \rightarrow \mathbb{Z}/2 \xrightarrow{0} \mathbb{Z}/2$.

B. Ring tensor products

(B1) Base change by a quotient: $(A/I) \otimes_A B \cong B/IB$. Example: $k[x]/(x) \otimes_{k[x]} k[x]/(x^2) \cong k$.

(B2) Disjoint fibers can give 0: $k[x]/(x) \otimes_{k[x]} k[x]/(x-1) \cong 0$.

(B3) Axes intersection: $k[x, y]/(x) \otimes_{k[x, y]} k[x, y]/(y) \cong k$.

(B4) Problem 4(a) pattern: $k \otimes_{k[x, y]} k[u, v] \cong k[u, v]/(u, uv) \cong k[v]$.

(B5) Problem 4(b) pattern: $k[v] \otimes_{k[x]} k[w] \cong k[v, w]/(v^2 - w^2)$; if $\text{char}(k) \neq 2$, then $\cong k[v] \times k[w]$; if $\text{char}(k) = 2$, then $v^2 - w^2 = (v - w)^2$ is nonreduced.

C. Flatness examples

(C1) $A[x]$ is free as an A -module: $A[x] \cong \bigoplus_{n \geq 0} A \cdot x^n$, hence flat.

(C2) Localization is flat: $S^{-1}A \otimes_A M \cong S^{-1}M$, and exact sequences remain exact after localizing.

(C3) $\mathbb{Q}$ is flat over $\mathbb{Z}$ since $\mathbb{Q} = \mathbb{Z}[S^{-1}]$ with $S = \mathbb{Z} \setminus \{0\}$.

(C4) Non-flat: $k[x]/(x^2)$ is not flat over $k[x]$ (tensoring can break injectivity).

D. Localization identities

(D1) Iterated localization: for $S = \{2^n\}$, $T = \{3^n\} \subset \mathbb{Z}$, $S^{-1}(T^{-1}\mathbb{Z}) \cong (ST)^{-1}\mathbb{Z} \cong \mathbb{Z}[1/6]$.

E. Quadric chart (Problem 7)

Let $R = k[x, y, z]/(xy - z^2)$. In R_x , x is invertible and $y = z^2/x$, hence $R_x \cong k[x, z]_x$. Similarly $R_y \cong k[y, z]_y$.

Element calculations for A1–A3

A1. $\mathbb{Z}^r \otimes_{\mathbb{Z}} M \cong M^r$

Let $e_1, \dots, e_r$ be the standard basis of $\mathbb{Z}^r$. Then for $v = (a_1, \dots, a_r) \in \mathbb{Z}^r$,

$$v \otimes m = \left(\sum_{i=1}^r a_i e_i \right) \otimes m = \sum_{i=1}^r e_i \otimes (a_i m).$$

Define $\Phi : \mathbb{Z}^r \otimes M \rightarrow M^r$ by

$$\Phi(e_i \otimes m) = (0, \dots, 0, m, 0, \dots, 0),$$

and extend linearly. Then

$$\Phi((a_1, \dots, a_r) \otimes m) = (a_1 m, \dots, a_r m).$$

The inverse is $\Psi : M^r \rightarrow \mathbb{Z}^r \otimes M$,

$$\Psi(m_1, \dots, m_r) = \sum_{i=1}^r e_i \otimes m_i.$$

A2. $(\mathbb{Z}/m) \otimes_{\mathbb{Z}} M \cong M/mM$

In $(\mathbb{Z}/m) \otimes M$, one has $\bar{a} \otimes x = \bar{1} \otimes (ax)$, and since $m\bar{1} = 0$,

$$0 = (m\bar{1}) \otimes x = \bar{1} \otimes (mx),$$

so mM maps to 0. Define

$$\Phi : (\mathbb{Z}/m) \otimes M \rightarrow M/mM, \quad \Phi(\bar{a} \otimes x) = ax + mM.$$

Its inverse is $\Psi : M/mM \rightarrow (\mathbb{Z}/m) \otimes M$,

$$\Psi(x + mM) = \bar{1} \otimes x.$$

A3. $(\mathbb{Z}/m) \otimes (\mathbb{Z}/n) \cong \mathbb{Z}/\gcd(m, n)$

Let $u = \bar{1} \in \mathbb{Z}/m$, $v = \bar{1} \in \mathbb{Z}/n$, and $t = u \otimes v$. Then $mt = 0$ and $nt = 0$, hence $gt = 0$ for $g = \gcd(m, n)$. Moreover,

$$\bar{a} \otimes \bar{b} = (au) \otimes (bv) = ab(u \otimes v) = abt,$$

so the tensor product is cyclic generated by t . Define

$$\Psi : (\mathbb{Z}/m) \otimes (\mathbb{Z}/n) \rightarrow \mathbb{Z}/g, \quad \Psi(\bar{a} \otimes \bar{b}) = \overline{ab} \pmod{g},$$

and $\Phi : \mathbb{Z}/g \rightarrow (\mathbb{Z}/m) \otimes (\mathbb{Z}/n)$ by $\Phi(\bar{1}) = t$. Then Φ and Ψ are inverse isomorphisms.

Element-wise view: the tensors and the basis vectors

e_i

What is e_i in $\mathbb{Z}^r$?

$$\mathbb{Z}^r = \{(a_1, \dots, a_r) \mid a_i \in \mathbb{Z}\}, \quad e_i = (0, \dots, 0, 1, 0, \dots, 0).$$

Every $v = (a_1, \dots, a_r)$ decomposes as $v = \sum_{i=1}^r a_i e_i$. Under $\mathbb{Z}^r \otimes_{\mathbb{Z}} M \cong M^r$, one has

$$e_i \otimes m \mapsto (0, \dots, 0, m, 0, \dots, 0).$$

Problem 4(a) element-wise. In $k \otimes_{k[x,y]} k[u, v]$, every element is $1 \otimes f(u, v)$. The tensor relations force $1 \otimes (uf) = 0$ because x acts as 0 on k but as u on $k[u, v]$. Hence $1 \otimes u = 0$, so the tensor product kills the ideal (u) and

$$k \otimes_{k[x,y]} k[u, v] \cong k[u, v]/(u) \cong k[v].$$

Example:

$$1 \otimes (u^2v + 3uv^5 + 7v^4) = 1 \otimes 7v^4.$$

$\varepsilon = v - w$ is nilpotent: $\varepsilon^2 = 0$.

Problem 4(b) element-wise. Rename variables: $k[v] \otimes_{k[x]} k[w]$ with $x \mapsto v^2$ and $x \mapsto w^2$. Then $v^2 \otimes 1 = 1 \otimes w^2$, and

$$k[v] \otimes_{k[x]} k[w] \cong k[v, w]/(v^2 - w^2).$$

A pure tensor $f(v) \otimes g(w)$ corresponds to the class of $f(v)g(w)$ modulo $(v^2 - w^2)$. If $\text{char}(k) \neq 2$, then $v^2 - w^2 = (v - w)(v + w)$ and CRT yields

$$k[v, w]/(v^2 - w^2) \cong k[v] \times k[v],$$

$$[F(v, w)] \mapsto (F(v, v), F(v, -v)).$$

If $\text{char}(k) = 2$, then $v^2 - w^2 = (v - w)^2$, so $\varepsilon := v - w$ satisfies $\varepsilon^2 = 0$.

What are elements in tensor products? (components and operations)

Module tensor $M \otimes_A N$

Elements are finite sums $\sum_j m_j \otimes n_j$ modulo the relations

$$(m + m') \otimes n = m \otimes n + m' \otimes n, \quad m \otimes (n + n') = m \otimes n + m \otimes n', \quad (am) \otimes n = m \otimes (an).$$

Addition is termwise; scalar multiplication is induced by $a(m \otimes n) = (am) \otimes n = m \otimes (an)$. If $M \cong A^r$ with basis e_i , then every element is $\sum_{i=1}^r e_i \otimes n_i$, i.e. corresponds to $(n_1, \dots, n_r) \in N^r$.

Ring tensor $B \otimes_A C$

As an abelian group it is formed the same way (finite sums $\sum b_i \otimes c_i$ modulo bilinear relations). It becomes a ring via

$$(b \otimes c)(b' \otimes c') = (bb') \otimes (cc'), \quad 1 = 1_B \otimes 1_C.$$

Balancing forces $(\varphi(a)b) \otimes c = b \otimes (\psi(a)c)$ for $a \in A$.

Example: $k \otimes_{k[x,y]} k[u, v]$

Every element is $1 \otimes f(u, v)$. Balancing with x, y gives $1 \otimes uf = 0$ for all f , hence $1 \otimes u = 0$, so the ring is $k[u, v]/(u) \cong k[v]$. Computation rule: $1 \otimes f(u, v) = 1 \otimes f(0, v)$.

Example: $k[v] \otimes_{k[x]} k[w]$ **with** $x \mapsto v^2, w^2$

Elements are sums $\sum f_i(v) \otimes g_i(w)$ and the key relation is $v^2 \otimes 1 = 1 \otimes w^2$. There is a presentation

$$k[v] \otimes_{k[x]} k[w] \cong k[v, w]/(v^2 - w^2),$$

sending $f(v) \otimes g(w) \mapsto [f(v)g(w)]$. Multiplication is induced from the quotient ring.

Problem 8 — Torsion submodule and its basic properties

Exact statement.

Let A be an integral domain and M an A -module.

An element $m \in M$ is *torsion* if there exists $a \in A \setminus \{0\}$ such that

$$am = 0.$$

Let $T(M)$ denote the set of all torsion elements of M .

Show that $T(M)$ is a submodule of M . If $T(M) = 0$, we say that M is *torsion-free*.

Show furthermore:

(a) If M is any A -module, then $M/T(M)$ is torsion-free.

(b) If $f : M \rightarrow N$ is a module homomorphism, then

$$f(T(M)) \subseteq T(N).$$

(c) If

$$0 \longrightarrow M_1 \xrightarrow{\alpha} M_2 \xrightarrow{\beta} M_3$$

is an exact sequence of A -modules, then so is

$$0 \longrightarrow T(M_1) \xrightarrow{\alpha} T(M_2) \xrightarrow{\beta} T(M_3).$$

(d) If $S \subseteq A$ is multiplicatively closed, show that

$$T(S^{-1}M) = S^{-1}T(M)$$

as $S^{-1}A$ -modules.

Solution.

Step 1: $T(M)$ is a submodule.

- $0 \in T(M)$ since for any $a \neq 0$ we have $a \cdot 0 = 0$.
- Let $m, n \in T(M)$. Then there exist $a, b \in A \setminus \{0\}$ such that

$$am = 0, \quad bn = 0.$$

Consider $ab \neq 0$. Then

$$(ab)(m + n) = abm + abn = b(am) + a(bn) = b \cdot 0 + a \cdot 0 = 0.$$

Hence $m + n \in T(M)$.

- Let $m \in T(M)$ and $r \in A$. Choose $a \neq 0$ with $am = 0$. Then

$$a(rm) = r(am) = r \cdot 0 = 0.$$

So $rm \in T(M)$.

Therefore $T(M) \leq M$.

(a) $M/T(M)$ is torsion-free.

Let $\bar{m} = m + T(M) \in M/T(M)$. Assume $\bar{m}$ is torsion, so there exists $a \neq 0$ such that

$$a\bar{m} = 0.$$

This means

$$am \in T(M).$$

So there exists $b \neq 0$ with

$$b(am) = 0.$$

Hence $(ba)m = 0$, with $ba \neq 0$. Therefore $m \in T(M)$, so $\bar{m} = 0$.

Thus $M/T(M)$ is torsion-free.

(b) Homomorphisms send torsion to torsion.

Let $m \in T(M)$. Then $\exists a \neq 0$ with $am = 0$. Apply f :

$$af(m) = f(am) = f(0) = 0.$$

So $f(m) \in T(N)$, and hence

$$f(T(M)) \subseteq T(N).$$

(c) Exactness of $0 \rightarrow T(M_1) \rightarrow T(M_2) \rightarrow T(M_3)$.

Assume

$$0 \longrightarrow M_1 \xrightarrow{\alpha} M_2 \xrightarrow{\beta} M_3$$

is exact.

- The restriction $\alpha : T(M_1) \rightarrow T(M_2)$ is injective because α is injective.
- We have $\beta \circ \alpha = 0$, hence

$$\beta(\alpha(T(M_1))) = 0,$$

so $\alpha(T(M_1)) \subseteq \ker(\beta|_{T(M_2)})$.

- Conversely, let $y \in T(M_2)$ with $\beta(y) = 0$. By exactness of $M_1 \rightarrow M_2 \rightarrow M_3$, there exists $x \in M_1$ such that

$$y = \alpha(x).$$

Since y is torsion, there exists $a \neq 0$ with $ay = 0$. Then

$$0 = ay = a\alpha(x) = \alpha(ax).$$

Because α is injective, $ax = 0$, so $x \in T(M_1)$. Therefore $y = \alpha(x) \in \alpha(T(M_1))$.

Hence

$$\ker(\beta|_{T(M_2)}) = \alpha(T(M_1)),$$

so the restricted sequence is exact.

(d) Localization commutes with torsion: $T(S^{-1}M) = S^{-1}T(M)$.

($\subseteq$) Let $\frac{m}{s} \in T(S^{-1}M)$. Then there exists $a \in A \setminus \{0\}$ such that

$$a \cdot \frac{m}{s} = \frac{am}{s} = 0 \quad \text{in } S^{-1}M.$$

By the criterion for a fraction to be zero, this means there exists $t \in S$ such that

$$t(am) = 0 \quad \text{in } M.$$

Since A is a domain, $ta \neq 0$. Hence $m \in T(M)$, and therefore

$$\frac{m}{s} \in S^{-1}T(M).$$

($\supseteq$) Let $\frac{m}{s} \in S^{-1}T(M)$. Then $m \in T(M)$, so there exists $a \neq 0$ with $am = 0$. Hence

$$a \cdot \frac{m}{s} = \frac{am}{s} = 0 \quad \text{in } S^{-1}M,$$

so $\frac{m}{s} \in T(S^{-1}M)$.

Therefore

$$T(S^{-1}M) = S^{-1}T(M)$$

as $S^{-1}A$ -modules.

Arithmetic examples: tensoring and adding in three equivalent views

Let $e_1 = (1, 0, 0)$, $e_2 = (0, 1, 0)$, $e_3 = (0, 0, 1)$ in $\mathbb{Z}^3$. Let $u = (1, 1, 1)$, $v = (2, -1, 3)$, $w = (0, 4, -2)$.

1) Pure-tensor / basis view

Since $u = e_1 + e_2 + e_3$,

$$e_2 \otimes u = e_2 \otimes (e_1 + e_2 + e_3) = e_2 \otimes e_1 + e_2 \otimes e_2 + e_2 \otimes e_3.$$

Compute $v + w = (2, 3, 1) = 2e_1 + 3e_2 + e_3$. Then

$$\begin{aligned} (e_1 + 2e_3) \otimes (v + w) &= e_1 \otimes (v + w) + 2e_3 \otimes (v + w) \\ &= 2(e_1 \otimes e_1) + 3(e_1 \otimes e_2) + (e_1 \otimes e_3) + 4(e_3 \otimes e_1) + 6(e_3 \otimes e_2) + 2(e_3 \otimes e_3). \end{aligned}$$

Also,

$$e_1 \otimes u + e_1 \otimes v = e_1 \otimes (u + v), \quad u + v = (3, 0, 4),$$

so

$$e_1 \otimes (u + v) = 3(e_1 \otimes e_1) + 4(e_1 \otimes e_3).$$

2) Tuple-of-vectors view $\mathbb{Z}^3 \otimes \mathbb{Z}^3 \cong (\mathbb{Z}^3)^3$

Under Φ , $\Phi(e_1 \otimes x) = (x, 0, 0)$, $\Phi(e_2 \otimes x) = (0, x, 0)$, $\Phi(e_3 \otimes x) = (0, 0, x)$. Hence

$$\Phi(e_2 \otimes u) = (0, u, 0) = ((0, 0, 0), (1, 1, 1), (0, 0, 0)).$$

Moreover,

$$\Phi((e_1 + 2e_3) \otimes (v + w)) = (v + w, 0, 2(v + w)) = ((2, 3, 1), (0, 0, 0), (4, 6, 2)).$$

Addition is componentwise:

$$(u, 0, 0) + (v, 0, 0) = (u + v, 0, 0).$$

3) Matrix (outer product) view $\mathbb{Z}^3 \otimes \mathbb{Z}^3 \cong \text{Mat}_{3 \times 3}(\mathbb{Z})$

Identify $a \otimes b$ with $ab^\top$. Then

$$e_2 \otimes u \leftrightarrow \begin{pmatrix} 0 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

Also, since $e_1 + 2e_3 = (1, 0, 2)$ and $v + w = (2, 3, 1)$,

$$(e_1 + 2e_3) \otimes (v + w) \leftrightarrow \begin{pmatrix} 2 & 3 & 1 \\ 0 & 0 & 0 \\ 4 & 6 & 2 \end{pmatrix}.$$

Finally,

$$e_1 \otimes u + e_1 \otimes v \leftrightarrow \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 2 & -1 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 3 & 0 & 4 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Arithmetic examples mod 3: tensoring and adding in three views

Work over $\mathbb{F}_3 = \mathbb{Z}/3$. Let

$$e_1 = (1, 0, 0), \quad e_2 = (0, 1, 0), \quad e_3 = (0, 0, 1)$$

in $\mathbb{F}_3^3$. Take

$$u = (1, 1, 1), \quad v = (2, -1, 3) \equiv (2, 2, 0), \quad w = (0, 4, -2) \equiv (0, 1, 1).$$

Then

$$v + w \equiv (2, 2, 0) + (0, 1, 1) = (2, 0, 1), \quad u + v \equiv (1, 1, 1) + (2, 2, 0) = (0, 0, 1).$$

1) Pure-tensor / basis view

Since $u = e_1 + e_2 + e_3$,

$$e_2 \otimes u = e_2 \otimes (e_1 + e_2 + e_3) = e_2 \otimes e_1 + e_2 \otimes e_2 + e_2 \otimes e_3.$$

Also $v + w = (2, 0, 1) = 2e_1 + e_3$, hence

$$(e_1 + 2e_3) \otimes (v + w) = e_1 \otimes (v + w) + 2e_3 \otimes (v + w).$$

Compute

$$e_1 \otimes (v + w) = 2(e_1 \otimes e_1) + (e_1 \otimes e_3),$$

$$2e_3 \otimes (v + w) = 4(e_3 \otimes e_1) + 2(e_3 \otimes e_3) \equiv (e_3 \otimes e_1) + 2(e_3 \otimes e_3),$$

so

$$(e_1 + 2e_3) \otimes (v + w) \equiv 2(e_1 \otimes e_1) + (e_1 \otimes e_3) + (e_3 \otimes e_1) + 2(e_3 \otimes e_3).$$

Finally,

$$e_1 \otimes u + e_1 \otimes v = e_1 \otimes (u + v) \equiv e_1 \otimes e_3.$$

2) Tuple-of-vectors view $\mathbb{F}_3^3 \otimes \mathbb{F}_3^3 \cong (\mathbb{F}_3^3)^3$

Under Φ , $\Phi(e_1 \otimes x) = (x, 0, 0)$, $\Phi(e_2 \otimes x) = (0, x, 0)$, $\Phi(e_3 \otimes x) = (0, 0, x)$. Thus

$$\Phi(e_2 \otimes u) = (0, u, 0) = ((0, 0, 0), (1, 1, 1), (0, 0, 0)),$$

and since $2(v + w) \equiv (1, 0, 2)$,

$$\Phi((e_1 + 2e_3) \otimes (v + w)) = (v + w, 0, 2(v + w)) = ((2, 0, 1), (0, 0, 0), (1, 0, 2)).$$

Addition is componentwise, e.g.

$$(u, 0, 0) + (v, 0, 0) = (u + v, 0, 0) = ((0, 0, 1), 0, 0).$$

3) Matrix (outer product) view over $\mathbb{F}_3$

Identify $a \otimes b$ with $ab^T \in \text{Mat}_{3 \times 3}(\mathbb{F}_3)$. Then

$$e_2 \otimes u \leftrightarrow \begin{pmatrix} 0 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad (e_1 + 2e_3) \otimes (v + w) \leftrightarrow \begin{pmatrix} 2 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 2 \end{pmatrix}.$$

Also

$$e_1 \otimes u \leftrightarrow \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad e_1 \otimes v \leftrightarrow \begin{pmatrix} 2 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

so

$$e_1 \otimes u + e_1 \otimes v \leftrightarrow \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

which corresponds to $e_1 \otimes e_3$.

Arithmetic examples mod 3: tensoring and adding in three views

Work over $\mathbb{F}_3 = \mathbb{Z}/3$. Let

$$e_1 = (1, 0, 0), \quad e_2 = (0, 1, 0), \quad e_3 = (0, 0, 1)$$

in $\mathbb{F}_3^3$. Take

$$u = (1, 1, 1), \quad v = (2, -1, 3) \equiv (2, 2, 0), \quad w = (0, 4, -2) \equiv (0, 1, 1).$$

Then

$$v + w \equiv (2, 2, 0) + (0, 1, 1) = (2, 0, 1), \quad u + v \equiv (1, 1, 1) + (2, 2, 0) = (0, 0, 1).$$

1) Pure-tensor / basis view

Since $u = e_1 + e_2 + e_3$,

$$e_2 \otimes u = e_2 \otimes (e_1 + e_2 + e_3) = e_2 \otimes e_1 + e_2 \otimes e_2 + e_2 \otimes e_3.$$

Also $v + w = (2, 0, 1) = 2e_1 + e_3$, hence

$$(e_1 + 2e_3) \otimes (v + w) = e_1 \otimes (v + w) + 2e_3 \otimes (v + w).$$

Compute

$$\begin{aligned} e_1 \otimes (v + w) &= 2(e_1 \otimes e_1) + (e_1 \otimes e_3), \\ 2e_3 \otimes (v + w) &= 4(e_3 \otimes e_1) + 2(e_3 \otimes e_3) \equiv (e_3 \otimes e_1) + 2(e_3 \otimes e_3), \end{aligned}$$

so

$$(e_1 + 2e_3) \otimes (v + w) \equiv 2(e_1 \otimes e_1) + (e_1 \otimes e_3) + (e_3 \otimes e_1) + 2(e_3 \otimes e_3).$$

Finally,

$$e_1 \otimes u + e_1 \otimes v = e_1 \otimes (u + v) \equiv e_1 \otimes e_3.$$

2) Tuple-of-vectors view $\mathbb{F}_3^3 \otimes \mathbb{F}_3^3 \cong (\mathbb{F}_3^3)^3$

Under Φ , $\Phi(e_1 \otimes x) = (x, 0, 0)$, $\Phi(e_2 \otimes x) = (0, x, 0)$, $\Phi(e_3 \otimes x) = (0, 0, x)$. Thus

$$\Phi(e_2 \otimes u) = (0, u, 0) = ((0, 0, 0), (1, 1, 1), (0, 0, 0)),$$

and since $2(v + w) \equiv (1, 0, 2)$,

$$\Phi((e_1 + 2e_3) \otimes (v + w)) = (v + w, 0, 2(v + w)) = ((2, 0, 1), (0, 0, 0), (1, 0, 2)).$$

Addition is componentwise, e.g.

$$(u, 0, 0) + (v, 0, 0) = (u + v, 0, 0) = ((0, 0, 1), 0, 0).$$

3) Matrix (outer product) view over $\mathbb{F}_3$

Identify $a \otimes b$ with $ab^\top \in \text{Mat}_{3 \times 3}(\mathbb{F}_3)$. Then

$$e_2 \otimes u \leftrightarrow \begin{pmatrix} 0 & 0 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}, \quad (e_1 + 2e_3) \otimes (v + w) \leftrightarrow \begin{pmatrix} 2 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 2 \end{pmatrix}.$$

Also

$$e_1 \otimes u \leftrightarrow \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad e_1 \otimes v \leftrightarrow \begin{pmatrix} 2 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

so

$$e_1 \otimes u + e_1 \otimes v \leftrightarrow \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

which corresponds to $e_1 \otimes e_3$.

Concrete tensoring on vectors (mod 3)

Let $k = \mathbb{F}_3$, and $V = W = \mathbb{F}_3^3$ with basis $e_1 = (1, 0, 0)$, $e_2 = (0, 1, 0)$, $e_3 = (0, 0, 1)$. Then $\{e_i \otimes e_j\}_{i,j}$ is a basis of $V \otimes_k W$, so any element is

$$T = \sum_{i,j} a_{ij}(e_i \otimes e_j), \quad a_{ij} \in \mathbb{F}_3,$$

and addition/scalars are coefficientwise.

If $a = \sum_i \alpha_i e_i$ and $b = \sum_j \beta_j e_j$, then

$$a \otimes b = \sum_{i,j} (\alpha_i \beta_j)(e_i \otimes e_j).$$

Equivalently $V \otimes_k W \cong \text{Mat}_{3 \times 3}(\mathbb{F}_3)$, with $a \otimes b \leftrightarrow ab^\top$.

Example: $a = (1, 0, 2) = e_1 + 2e_3$, $b = (2, 0, 1) = 2e_1 + e_3$. Then

$$a \otimes b = 2(e_1 \otimes e_1) + (e_1 \otimes e_3) + (e_3 \otimes e_1) + 2(e_3 \otimes e_3),$$

since $4 \equiv 1 \pmod{3}$. In matrix form,

$$ab^\top \equiv \begin{pmatrix} 2 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 2 \end{pmatrix}.$$

Theory — k -algebras, tensor products, and concrete vector tensors

1) Theory: what is a k -algebra?

A k -algebra is a ring B together with a ring homomorphism

$$\iota : k \longrightarrow B$$

whose image lies in the center of B (automatic if B is commutative).

Intuitively:

B is a ring where you can multiply by scalars from k .

Examples.

- $k[x]$, $k[x, y]$, $k[u, v]$ are k -algebras.
- Any quotient $k[x, y]/I$ is a k -algebra.
- Any localization $S^{-1}k[x]$ is a k -algebra.

2) Tensoring k -algebras: what is $B \otimes_k C$?

If B and C are commutative k -algebras, then $B \otimes_k C$ is:

2.1 Elements

Elements are finite sums of pure tensors

$$\sum_{i=1}^t b_i \otimes c_i,$$

modulo the bilinear relations (over k):

$$\begin{aligned} (b + b') \otimes c &= b \otimes c + b' \otimes c, \\ b \otimes (c + c') &= b \otimes c + b \otimes c', \\ (\lambda b) \otimes c &= b \otimes (\lambda c) \quad (\lambda \in k). \end{aligned}$$

2.2 Ring operations

Addition is termwise.

Multiplication is defined on pure tensors by

$$(b \otimes c)(b' \otimes c') = (bb') \otimes (cc')$$

and extended by distributivity.

The unit is

$$1_{B \otimes_k C} = 1_B \otimes 1_C.$$

2.3 Universal property (the meaning)

$B \otimes_k C$ is the most general k -algebra receiving k -algebra maps from both B and C whose images commute.

Geometric slogan:

$$\mathrm{Spec}(B \otimes_k C) \cong \mathrm{Spec}(B) \times_{\mathrm{Spec}(k)} \mathrm{Spec}(C).$$

3) Tensoring over a base algebra A : $B \otimes_A C$

Now assume $A \rightarrow B$ and $A \rightarrow C$ are k -algebra maps. Then $B \otimes_A C$ is the same construction, but with an extra identification.

3.1 The balancing relation

For $a \in A$, $b \in B$, $c \in C$,

$$(\varphi(a)b) \otimes c = b \otimes (\psi(a)c),$$

where $\varphi : A \rightarrow B$ and $\psi : A \rightarrow C$.

Interpretation:

the two copies of A become the same inside the tensor.

3.2 Presentation rules (fast computation)

Rule 1 (base change by a quotient). If $A \rightarrow B$ and $I \triangleleft A$, then

$$(A/I) \otimes_A B \cong B/IB.$$

This means: impose the relations defining A/I inside B .

Rule 2 (polynomial presentation). If $B \cong A[u]/J$ and $C \cong A[v]/K$, then

$$B \otimes_A C \cong A[u, v]/(J + K + (\text{compatibility relations})).$$

In practice: rename variables, then add relations forcing images of generators of A to match.

4) Your algebra examples: element-wise meaning

4(a) $k \otimes_{k[x, y]} k[u, v]$

Every element can be written as

$$1 \otimes f(u, v).$$

Here x, y act as 0 on k , but act as u, uv on $k[u, v]$. Balancing forces

$$1 \otimes (uf) = 0 \quad \forall f,$$

so in particular $1 \otimes u = 0$. Thus u is killed and

$$k \otimes_{k[x, y]} k[u, v] \cong k[u, v]/(u) \cong k[v].$$

Computation rule:

$$1 \otimes f(u, v) = 1 \otimes f(0, v).$$

4(b) $k[v] \otimes_{k[x]} k[w]$ with $x \mapsto v^2$, $x \mapsto w^2$

Elements are sums

$$\sum f_i(v) \otimes g_i(w).$$

Key balancing relation:

$$v^2 \otimes 1 = 1 \otimes w^2.$$

Presentation:

$$k[v] \otimes_{k[x]} k[w] \cong k[v, w]/(v^2 - w^2).$$

Computation rule: in the quotient, replace v^2 with w^2 (and vice versa), because $v^2 - w^2 = 0$.

5) Tensoring on concrete vectors (the $\mathbb{F}_3^3$ case)

Now take $k = \mathbb{F}_3 = \mathbb{Z}/3$. Then $V = \mathbb{F}_3^3$ is a k -vector space with basis

$$e_1 = (1, 0, 0), \quad e_2 = (0, 1, 0), \quad e_3 = (0, 0, 1).$$

Then $\{e_i \otimes e_j\}_{i,j}$ is a basis of $V \otimes_k V$, so

$$\dim_k(V \otimes_k V) = 9.$$

5.1 What are elements?

Elements are finite sums

$$T = \sum_{i,j} a_{ij} (e_i \otimes e_j), \quad a_{ij} \in \mathbb{F}_3.$$

So T is determined by a 3×3 grid of coefficients (a_{ij}) .

5.2 Operations

Addition:

$$\sum a_{ij} e_i \otimes e_j + \sum b_{ij} e_i \otimes e_j = \sum (a_{ij} + b_{ij}) e_i \otimes e_j,$$

with all coefficients computed mod 3.

Scalar multiplication:

$$\lambda \cdot \sum a_{ij} e_i \otimes e_j = \sum (\lambda a_{ij}) e_i \otimes e_j \quad (\lambda \in \mathbb{F}_3).$$

5.3 Pure tensors and bilinearity (how you compute)

If

$$a = \sum_i \alpha_i e_i, \quad b = \sum_j \beta_j e_j,$$

then

$$a \otimes b = \sum_{i,j} (\alpha_i \beta_j) (e_i \otimes e_j).$$

This is the outer-product coefficient rule.

5.4 Matrix identification (outer product)

There is a natural identification

$$\mathbb{F}_3^3 \otimes_{\mathbb{F}_3} \mathbb{F}_3^3 \cong \text{Mat}_{3 \times 3}(\mathbb{F}_3),$$

sending $e_i \otimes e_j \mapsto E_{ij}$. Then a pure tensor $a \otimes b$ corresponds to the rank-one matrix ab^T .

Concrete worked vector example (mod 3).

Take

$$a = (1, 0, 2) = e_1 + 2e_3, \quad b = (2, 0, 1) = 2e_1 + e_3.$$

Then

$$a \otimes b = (e_1 + 2e_3) \otimes (2e_1 + e_3) = 2(e_1 \otimes e_1) + (e_1 \otimes e_3) + 4(e_3 \otimes e_1) + 2(e_3 \otimes e_3).$$

Since $4 \equiv 1 \pmod{3}$, we get

$$a \otimes b = 2(e_1 \otimes e_1) + (e_1 \otimes e_3) + (e_3 \otimes e_1) + 2(e_3 \otimes e_3).$$

Matrix form:

$$ab^T = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} \begin{pmatrix} 2 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 & 1 \\ 0 & 0 & 0 \\ 4 & 0 & 2 \end{pmatrix} \equiv \begin{pmatrix} 2 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 2 \end{pmatrix} \pmod{3}.$$

Problem 9 — Any n generators of A^n form a basis

Exact statement.

Let A be a ring.

Let F be the free A -module

$$F = A^n.$$

Show that every set of n generators

$$x_1, \dots, x_n$$

of F is a basis of F .

Equivalently, if

$$\sum_{i=1}^n a_i x_i = 0,$$

then

$$a_i = 0 \quad \text{for all } i.$$

Solution.

Define an A -linear map

$$\varphi : A^n \longrightarrow F$$

by

$$\varphi(e_i) = x_i,$$

where $e_1, \dots, e_n$ is the standard basis of A^n .

Since $x_1, \dots, x_n$ generate F , the map φ is surjective.

We must show that φ is injective.

Step 1: localize at a maximal ideal.

Let $\mathfrak{m}$ be a maximal ideal of A .

Localizing gives a surjective map

$$\varphi_{\mathfrak{m}} : A_{\mathfrak{m}}^n \longrightarrow A_{\mathfrak{m}}^n.$$

So it is enough to prove:

If R is a local ring, then every surjective endomorphism

$$R^n \longrightarrow R^n$$

is an isomorphism.

Take $R = A_{\mathfrak{m}}$, with maximal ideal still denoted $\mathfrak{m}$.

Step 2: reduce modulo the maximal ideal.

Reduce $\varphi_{\mathfrak{m}}$ modulo $\mathfrak{m}$.

This gives a linear map of vector spaces over the residue field

$$k = R/\mathfrak{m} :$$

$$\overline{\varphi} : k^n \longrightarrow k^n.$$

Because $\varphi_{\mathfrak{m}}$ is surjective, the map $\overline{\varphi}$ is also surjective.

A surjective linear map

$$k^n \longrightarrow k^n$$

is automatically bijective.

Hence $\overline{\varphi}$ is an isomorphism.

Step 3: use the determinant.

Choose the matrix $M \in \text{Mat}_{n \times n}(R)$ representing $\varphi_{\mathfrak{m}}$.

Since $\overline{\varphi}$ is an isomorphism, its matrix $\overline{M}$ over k has nonzero determinant:

$$\det(\overline{M}) \neq 0.$$

Therefore

$$\det(M) \notin \mathfrak{m}.$$

Since R is local, every element outside $\mathfrak{m}$ is a unit.

So $\det(M)$ is a unit in R .

Hence M is invertible, and therefore

$$\varphi_{\mathfrak{m}} : R^n \longrightarrow R^n$$

is an isomorphism.

Thus $\varphi_{\mathfrak{m}}$ is injective for every maximal ideal $\mathfrak{m}$.

Step 4: conclude globally.

A map between finite free A -modules is injective if and only if all of its localizations at maximal ideals are injective.

Therefore φ is injective.

Since φ is both surjective and injective, it is an isomorphism.

Hence

$$x_1, \dots, x_n$$

is a basis of $F = A^n$.

Problem 10 — The Zariski topology on $\text{Spec } A$

Exact statement.

Let A be a ring.

Define

$$\text{Spec } A := \{\mathfrak{p} \subseteq A \mid \mathfrak{p} \text{ is a prime ideal}\}.$$

For an ideal $I \subseteq A$, define

$$V(I) := \{\mathfrak{p} \in \operatorname{Spec} A \mid I \subseteq \mathfrak{p}\}.$$

Show the following.

- (a) The sets $V(I)$ form the closed sets of a topology on $\operatorname{Spec} A$.

This topology is called the *Zariski topology*.

- (b) For $f \in A$, define

$$D(f) := \{\mathfrak{p} \in \operatorname{Spec} A \mid f \notin \mathfrak{p}\}.$$

Show that the sets $D(f)$ form a basis for the topology on $\operatorname{Spec} A$.

- (c) If

$$\varphi : A \longrightarrow B$$

is a ring homomorphism, show that for every $\mathfrak{q} \in \operatorname{Spec} B$,

$$\varphi^{-1}(\mathfrak{q})$$

is a prime ideal of A .

Hence φ induces a map

$$\varphi^* : \operatorname{Spec} B \longrightarrow \operatorname{Spec} A, \quad \mathfrak{q} \longmapsto \varphi^{-1}(\mathfrak{q}).$$

Show that φ^* is continuous for the Zariski topology.

Solution.

(a) The sets $V(I)$ are the closed sets.

We check the axioms of closed sets.

Whole space and empty set.

Every prime ideal contains 0, so

$$V(0) = \operatorname{Spec} A.$$

No prime ideal contains 1, so

$$V((1)) = \emptyset.$$

Arbitrary intersections.

Let $\{I_\lambda\}_{\lambda \in \Lambda}$ be any family of ideals.

Then

$$\mathfrak{p} \in \bigcap_{\lambda} V(I_\lambda)$$

if and only if

$$I_\lambda \subseteq \mathfrak{p} \quad \text{for all } \lambda.$$

This is equivalent to

$$\sum_{\lambda} I_\lambda \subseteq \mathfrak{p}.$$

Therefore

$$\bigcap_{\lambda} V(I_{\lambda}) = V\left(\sum_{\lambda} I_{\lambda}\right).$$

Finite unions.

For ideals $I, J \subseteq A$, we claim

$$V(I) \cup V(J) = V(IJ).$$

Indeed, if $\mathfrak{p}$ is prime, then

$$IJ \subseteq \mathfrak{p}$$

if and only if

$$I \subseteq \mathfrak{p} \quad \text{or} \quad J \subseteq \mathfrak{p}.$$

Hence

$$V(I) \cup V(J) = V(IJ).$$

By induction, finite unions of sets of the form $V(I)$ are again of this form. Thus the sets $V(I)$ define a topology on $\text{Spec } A$.

(b) The sets $D(f)$ form a basis.

By definition,

$$D(f) = \text{Spec } A \setminus V((f)),$$

so each $D(f)$ is open.

They cover the space.

Since

$$D(1) = \text{Spec } A,$$

the sets $D(f)$ cover $\text{Spec } A$.

Intersections of basic opens.

If $\mathfrak{p} \in \text{Spec } A$, then

$$\mathfrak{p} \in D(f) \cap D(g)$$

if and only if

$$f \notin \mathfrak{p} \quad \text{and} \quad g \notin \mathfrak{p}.$$

Since $\mathfrak{p}$ is prime, this is equivalent to

$$fg \notin \mathfrak{p}.$$

Therefore

$$D(f) \cap D(g) = D(fg).$$

Every open set is a union of basic opens.

Let U be open.

Then

$$U = \text{Spec } A \setminus V(I)$$

for some ideal $I \subseteq A$.

Now

$$\mathfrak{p} \in U$$

if and only if

$$I \not\subseteq \mathfrak{p}.$$

So there exists some $f \in I$ such that

$$f \notin \mathfrak{p}.$$

Thus

$$\mathfrak{p} \in D(f) \quad \text{for some } f \in I.$$

Hence

$$U = \bigcup_{f \in I} D(f).$$

So the sets $D(f)$ form a basis.

(c) The induced map on spectra is continuous.

Let

$$\varphi : A \longrightarrow B$$

be a ring homomorphism.

Take $\mathfrak{q} \in \text{Spec } B$.

First, $\varphi^{-1}(\mathfrak{q})$ is an ideal of A .

If

$$ab \in \varphi^{-1}(\mathfrak{q}),$$

then

$$\varphi(ab) \in \mathfrak{q}.$$

That is,

$$\varphi(a)\varphi(b) \in \mathfrak{q}.$$

Since $\mathfrak{q}$ is prime, either

$$\varphi(a) \in \mathfrak{q} \quad \text{or} \quad \varphi(b) \in \mathfrak{q}.$$

Hence

$$a \in \varphi^{-1}(\mathfrak{q}) \quad \text{or} \quad b \in \varphi^{-1}(\mathfrak{q}).$$

So $\varphi^{-1}(\mathfrak{q})$ is a prime ideal of A .

Therefore φ induces a map

$$\varphi^* : \text{Spec } B \longrightarrow \text{Spec } A, \quad \mathfrak{q} \longmapsto \varphi^{-1}(\mathfrak{q}).$$

Continuity.

It is enough to check inverse images of the basic opens $D(f)$.

Let $f \in A$.

Then

$$(\varphi^*)^{-1}(D(f)) = \{\mathfrak{q} \in \text{Spec } B \mid \varphi^{-1}(\mathfrak{q}) \in D(f)\}.$$

Now

$$\varphi^{-1}(\mathfrak{q}) \in D(f)$$

if and only if

$$f \notin \varphi^{-1}(\mathfrak{q}),$$

which is equivalent to

$$\varphi(f) \notin \mathfrak{q}.$$

So

$$(\varphi^*)^{-1}(D(f)) = D(\varphi(f)).$$

This is open in $\text{Spec } B$.

Hence φ^* is continuous.

Problem 11 — Disconnected spectrum, orthogonal idempotents, and product decomposition

Exact statement.

Let A be a ring.

Show that the following are equivalent:

- (i) $\text{Spec } A$ is disconnected.
- (ii) There exist nonzero elements

$$e_1, e_2 \in A$$

such that

$$e_1 e_2 = 0, \quad e_1^2 = e_1, \quad e_2^2 = e_2, \quad e_1 + e_2 = 1.$$

- (iii) A is isomorphic to a product

$$A_1 \times A_2$$

for some nonzero rings A_1, A_2 .

Note. The word “nonzero” in (iii) is necessary; otherwise $A \cong A \times 0$ is a trivial decomposition.

Solution.

We prove

$$(i) \iff (ii) \iff (iii).$$

$$(ii) \implies (iii).$$

Assume there exist nonzero elements $e_1, e_2 \in A$ such that

$$e_1^2 = e_1, \quad e_2^2 = e_2, \quad e_1 e_2 = 0, \quad e_1 + e_2 = 1.$$

Set

$$A_1 := Ae_1, \quad A_2 := Ae_2.$$

Define

$$\Phi : A \longrightarrow A_1 \times A_2, \quad \Phi(a) = (ae_1, ae_2).$$

This is a ring homomorphism.

If $\Phi(a) = (0, 0)$, then

$$ae_1 = 0, \quad ae_2 = 0,$$

so

$$a = a(e_1 + e_2) = ae_1 + ae_2 = 0.$$

Thus Φ is injective.

Now let $(x, y) \in A_1 \times A_2$. Write

$$x = ae_1, \quad y = be_2$$

for some $a, b \in A$. Then

$$\Phi(ae_1 + be_2) = ((ae_1 + be_2)e_1, (ae_1 + be_2)e_2) = (ae_1, be_2) = (x, y),$$

since $e_1e_2 = e_2e_1 = 0$.

So Φ is surjective.

Hence

$$A \cong Ae_1 \times Ae_2.$$

Because $e_1 \neq 0$ and $e_2 \neq 0$, both factors are nonzero.

(iii) $\implies$ (ii).

Assume

$$A \cong A_1 \times A_2$$

with $A_1, A_2 \neq 0$.

In $A_1 \times A_2$, define

$$e_1 = (1, 0), \quad e_2 = (0, 1).$$

Then

$$e_1^2 = e_1, \quad e_2^2 = e_2, \quad e_1e_2 = 0, \quad e_1 + e_2 = (1, 1) = 1.$$

Also $e_1 \neq 0$ and $e_2 \neq 0$ because both factors are nonzero.

Transport these elements back to A through the isomorphism.

Thus (ii) holds.

(ii) $\implies$ (i).

Assume (ii).

Consider the basic open sets

$$D(e_1) = \{\mathfrak{p} \in \operatorname{Spec} A \mid e_1 \notin \mathfrak{p}\}, \quad D(e_2) = \{\mathfrak{p} \in \operatorname{Spec} A \mid e_2 \notin \mathfrak{p}\}.$$

We claim that

$$\operatorname{Spec} A = D(e_1) \sqcup D(e_2).$$

Let $\mathfrak{p} \in \operatorname{Spec} A$.

Since

$$e_1e_2 = 0 \in \mathfrak{p}$$

and $\mathfrak{p}$ is prime, we must have

$$e_1 \in \mathfrak{p} \quad \text{or} \quad e_2 \in \mathfrak{p}.$$

They cannot both lie in $\mathfrak{p}$, because then

$$1 = e_1 + e_2 \in \mathfrak{p},$$

which is impossible.

Therefore exactly one of e_1, e_2 lies in $\mathfrak{p}$, so exactly one of the conditions

$$e_1 \notin \mathfrak{p}, \quad e_2 \notin \mathfrak{p}$$

holds.

Hence

$$\operatorname{Spec} A = D(e_1) \sqcup D(e_2).$$

Now both sets are nonempty. Indeed, if $D(e_1) = \emptyset$, then e_1 lies in every prime ideal, hence e_1 is nilpotent. But $e_1^2 = e_1$, so an idempotent nilpotent element must be 0, contradiction. Similarly $D(e_2) \neq \emptyset$.

Thus $\operatorname{Spec} A$ is disconnected.

$(i) \implies (ii)$.

Assume $\operatorname{Spec} A$ is disconnected.

Then there exist disjoint nonempty subsets

$$U, V \subseteq \operatorname{Spec} A$$

which are both open and closed, and such that

$$\operatorname{Spec} A = U \sqcup V.$$

Because $\operatorname{Spec} A$ is quasi-compact, each clopen subset is quasi-compact open. Hence we may write

$$U = D(I), \quad V = D(J)$$

for some finitely generated ideals $I, J \subseteq A$.

Since

$$U \cup V = \operatorname{Spec} A,$$

we have

$$D(I) \cup D(J) = \operatorname{Spec} A.$$

But

$$D(I) \cup D(J) = D(I + J),$$

so

$$D(I + J) = \operatorname{Spec} A.$$

Therefore

$$V(I + J) = \emptyset,$$

which implies

$$I + J = A.$$

Since

$$U \cap V = \emptyset,$$

we have

$$D(I) \cap D(J) = \emptyset.$$

But

$$D(I) \cap D(J) = D(IJ),$$

so

$$D(IJ) = \emptyset.$$

Thus

$$V(IJ) = \text{Spec } A.$$

Hence every element of IJ is nilpotent.

Because IJ is finitely generated, it is a nilpotent ideal. So for some $N \geq 1$,

$$(IJ)^N = 0.$$

Since $I + J = A$, the powers are still comaximal:

$$I^N + J^N = A.$$

Choose

$$e_1 \in I^N, \quad e_2 \in J^N$$

such that

$$e_1 + e_2 = 1.$$

Then

$$e_1 e_2 \in I^N J^N = (IJ)^N = 0.$$

Therefore

$$e_1^2 = e_1(e_1 + e_2) = e_1, \quad e_2^2 = e_2(e_1 + e_2) = e_2.$$

So e_1, e_2 are orthogonal idempotents and

$$e_1 + e_2 = 1.$$

They are nonzero because U and V are nonempty. For instance, if $e_1 = 0$, then $e_2 = 1$, so

$$V = D(J) = D(e_2) = D(1) = \text{Spec } A,$$

contradicting $U \neq \emptyset$. Similarly $e_2 \neq 0$.

Thus (ii) holds.

Hence

$$(i) \iff (ii) \iff (iii).$$

1 Tensoring with A/I : detailed explanation of the four examples

The common principle behind all four examples is the canonical isomorphism

$$(A/I) \otimes_A M \cong M/IM,$$

where:

- A is a ring,
- $I \subseteq A$ is an ideal,
- M is an A -module,
- IM is the submodule generated by all products im with $i \in I$ and $m \in M$.

Intuitively, tensoring with A/I means that every element of the ideal I is forced to act as 0. In other words, we are “reducing the module M modulo I .”

1.1 Element-level meaning of $(A/I) \otimes_A M$

An element of $(A/I) \otimes_A M$ is a finite sum of pure tensors

$$(\bar{a}) \otimes m,$$

where $\bar{a} = a + I \in A/I$ and $m \in M$.

Because $\bar{a} = a \cdot \bar{1}$ in A/I , we have

$$(\bar{a}) \otimes m = (\bar{1}) \otimes am.$$

So every pure tensor can be rewritten in the form

$$\bar{1} \otimes m.$$

Now if $i \in I$, then $\bar{i} = 0$ in A/I , hence

$$\bar{1} \otimes (im) = \bar{i} \otimes m = 0.$$

Therefore every element of IM becomes zero in the tensor product.

This is exactly why the tensor product depends only on the class of m modulo IM . The isomorphism is:

$$\Phi : (A/I) \otimes_A M \longrightarrow M/IM, \quad (\bar{a}) \otimes m \longmapsto am + IM.$$

Its inverse is given by

$$\Psi : M/IM \longrightarrow (A/I) \otimes_A M, \quad m + IM \longmapsto \bar{1} \otimes m.$$

So the tensor product remembers precisely the part of M that survives after we impose the rule $I = 0$.

1.2 What survives and what dies

Under the passage from M to $(A/I) \otimes_A M$:

- terms not involving the ideal I survive,
- all terms in the submodule IM vanish.

Thus:

- over $\mathbb{Z}$, tensoring with $\mathbb{Z}/m$ means reducing modulo m ,
- over $k[x]$, tensoring with $k[x]/(x)$ means setting $x = 0$.

2 Example 1: $A = \mathbb{Z}$, $I = (m)$, $M = \mathbb{Z}^r$

We compute

$$(\mathbb{Z}/m) \otimes_{\mathbb{Z}} \mathbb{Z}^r.$$

By the general formula,

$$(\mathbb{Z}/m) \otimes_{\mathbb{Z}} \mathbb{Z}^r \cong \mathbb{Z}^r / m\mathbb{Z}^r.$$

Now

$$m\mathbb{Z}^r = \{(ma_1, \dots, ma_r) : a_1, \dots, a_r \in \mathbb{Z}\}.$$

So $\mathbb{Z}^r/m\mathbb{Z}^r$ is obtained by reducing each coordinate modulo m . Therefore

$$\mathbb{Z}^r/m\mathbb{Z}^r \cong (\mathbb{Z}/m)^r.$$

Hence

$$(\mathbb{Z}/m) \otimes_{\mathbb{Z}} \mathbb{Z}^r \cong (\mathbb{Z}/m)^r.$$

2.1 Element-wise description

Let $e_1, \dots, e_r$ be the standard basis of $\mathbb{Z}^r$. Any vector $v \in \mathbb{Z}^r$ can be written as

$$v = a_1 e_1 + \dots + a_r e_r.$$

Then

$$[1]_m \otimes v = [1]_m \otimes (a_1 e_1 + \dots + a_r e_r) = [a_1]_m \otimes e_1 + \dots + [a_r]_m \otimes e_r.$$

So the tensor product simply remembers the coefficients modulo m .

2.2 Concrete example

Take $m = 5$, $r = 2$, and $v = (7, 11)$. Then

$$(7, 11) \equiv (2, 1) \pmod{5}.$$

Thus

$$[1]_5 \otimes (7, 11) \text{ corresponds to } ([2]_5, [1]_5) \in (\mathbb{Z}/5)^2.$$

2.3 Interpretation

The “vectors” here are genuine integer vectors in $\mathbb{Z}^r$, but after tensoring with $\mathbb{Z}/m$, only their coordinates modulo m remain visible.

3 Example 2: $A = \mathbb{Z}$, $I = (m)$, $M = \mathbb{Z}/n$

We compute

$$(\mathbb{Z}/m) \otimes_{\mathbb{Z}} (\mathbb{Z}/n).$$

Using the same principle,

$$(\mathbb{Z}/m) \otimes_{\mathbb{Z}} (\mathbb{Z}/n) \cong (\mathbb{Z}/n)/m(\mathbb{Z}/n).$$

So we must understand the subgroup $m(\mathbb{Z}/n)$.

3.1 Computing $m(\mathbb{Z}/n)$

Inside $\mathbb{Z}/n$, multiplication by m gives the subgroup

$$m(\mathbb{Z}/n) = \{[mk]_n : k \in \mathbb{Z}\}.$$

This is the cyclic subgroup generated by $[m]_n$.

The order of $[m]_n$ in $\mathbb{Z}/n$ is

$$\frac{n}{\gcd(m, n)},$$

so the subgroup $m(\mathbb{Z}/n)$ has exactly that many elements.

Therefore the quotient $(\mathbb{Z}/n)/m(\mathbb{Z}/n)$ has cardinality

$$\frac{n}{n/\gcd(m, n)} = \gcd(m, n).$$

Since the quotient of a cyclic group is cyclic, we get

$$(\mathbb{Z}/n)/m(\mathbb{Z}/n) \cong \mathbb{Z}/\gcd(m, n).$$

Hence

$$(\mathbb{Z}/m) \otimes_{\mathbb{Z}} (\mathbb{Z}/n) \cong \mathbb{Z}/\gcd(m, n).$$

3.2 Element-wise description

The tensor product is generated by the single element

$$[1]_m \otimes [1]_n.$$

Indeed, for any $a, b \in \mathbb{Z}$,

$$[a]_m \otimes [b]_n = ab([1]_m \otimes [1]_n).$$

So the whole tensor product is cyclic.

Moreover,

$$m([1]_m \otimes [1]_n) = 0 \quad \text{and} \quad n([1]_m \otimes [1]_n) = 0,$$

so the order of the generator divides both m and n , hence divides $\gcd(m, n)$. In fact, its exact order is $\gcd(m, n)$.

3.3 Concrete example

Let $m = 12$ and $n = 18$. Then

$$\gcd(12, 18) = 6,$$

so

$$(\mathbb{Z}/12) \otimes_{\mathbb{Z}} (\mathbb{Z}/18) \cong \mathbb{Z}/6.$$

The element

$$[1]_{12} \otimes [1]_{18}$$

is a generator of this cyclic group.

3.4 Interpretation

This tensor product retains only the *common torsion* visible in both cyclic modules:

- $\mathbb{Z}/m$ has m -torsion,
- $\mathbb{Z}/n$ has n -torsion,
- their tensor product detects the part compatible with both, namely $\gcd(m, n)$ -torsion.

4 Example 3: $A = k[x]$, $I = (x)$ — evaluation at $x = 0$

Now let

$$A = k[x], \quad I = (x).$$

Then

$$A/I = k[x]/(x).$$

But quotienting by (x) kills all multiples of x , so every polynomial becomes equal to its constant term. Thus

$$k[x]/(x) \cong k,$$

via the map

$$f(x) \mapsto f(0).$$

So in this situation,

$$A/I \cong k.$$

Therefore for any A -module M ,

$$k \otimes_{k[x]} M \cong M/xM.$$

4.1 Why this means “set $x = 0$ ”

Since $x = 0$ in the quotient ring $k[x]/(x)$, we have in the tensor product:

$$1 \otimes (xm) = x \otimes m = 0.$$

So every element of the form xm vanishes. That is why we divide by xM .

This is exactly the module-theoretic meaning of evaluating at $x = 0$: all x -terms disappear.

4.2 Particular case: $M = k[x]$

If $M = A = k[x]$, then

$$k \otimes_{k[x]} k[x] \cong k[x]/xk[x].$$

But $xk[x]$ is the set of polynomials with zero constant term, so the quotient keeps only the constant term:

$$k[x]/xk[x] \cong k.$$

Thus

$$k \otimes_{k[x]} k[x] \cong k.$$

4.3 Element-wise example

Take the polynomial

$$f(x) = 3 + 2x + x^2.$$

Then in the tensor product,

$$1 \otimes f(x) = 1 \otimes (3 + 2x + x^2).$$

Since all terms involving x vanish,

$$1 \otimes (3 + 2x + x^2) = 1 \otimes 3.$$

So only the constant term remains.

4.4 Interpretation

Tensoring with $k \cong k[x]/(x)$ means:

- forget all multiples of x ,
- keep only what survives at $x = 0$,
- algebraically: pass from M to the quotient M/xM .

5 Example 4: $A = k[x]$, $I = (x)$, $M = A/(x^2)$

Now let

$$A = k[x], \quad I = (x), \quad M = A/(x^2) = k[x]/(x^2).$$

We want to compute

$$(A/I) \otimes_A M = k \otimes_{k[x]} A/(x^2).$$

By the general rule,

$$(A/I) \otimes_A M \cong M/xM.$$

So we must compute xM .

5.1 Structure of $M = A/(x^2)$

In the quotient $k[x]/(x^2)$, every polynomial is equivalent to one of the form

$$a + bx \pmod{x^2}.$$

So as a k -vector space,

$$M = k \cdot 1 \oplus k \cdot x.$$

A convenient basis is

$$\{1, x\}.$$

However, inside this module we have

$$x^2 = 0.$$

So x is not zero, but it is nilpotent.

5.2 Computing xM

Take an arbitrary element

$$a + bx \in M.$$

Multiplying by x , we get

$$x(a + bx) = ax + bx^2.$$

Since $x^2 = 0$ in M , this becomes

$$x(a + bx) = ax.$$

Therefore

$$xM = kx.$$

So the quotient is

$$M/xM = (k \cdot 1 \oplus k \cdot x)/(k \cdot x) \cong k.$$

Hence

$$k \otimes_{k[x]} A/(x^2) \cong M/xM \cong k.$$

5.3 Element-wise tensor computation

Take the class of $a + bx$ in $M = A/(x^2)$. Then in the tensor product,

$$1 \otimes (a + bx) = 1 \otimes a + 1 \otimes bx.$$

But

$$1 \otimes bx = b(1 \otimes x).$$

Now $x = 0$ in $k = A/(x)$, so

$$1 \otimes x = x \otimes 1 = 0.$$

Therefore

$$1 \otimes (a + bx) = 1 \otimes a.$$

So again, only the constant part survives.

5.4 Interpretation

This example is important because $M = A/(x^2)$ is larger than k : it has an extra “infinitesimal” direction kx . But after tensoring with $k = A/(x)$, the x -part disappears.

So even though $x \neq 0$ in M , it becomes invisible after imposing $x = 0$.

6 What are the “vectors” or “elements” in these examples?

These are not always vectors in the geometric sense. They are simply *elements of modules*.

- In Example 1, the elements are tuples

$$(a_1, \dots, a_r) \in \mathbb{Z}^r.$$

These behave like integer vectors.

- In Example 2, the elements are residue classes

$$[a]_n \in \mathbb{Z}/n.$$

- In Example 3, the elements are arbitrary module elements $m \in M$, where M is any $k[x]$ -module.
- In Example 4, the elements are classes

$$a + bx \pmod{x^2},$$

so each element has a constant part and an x -part.

In all cases, after tensoring with A/I , every element can be represented by something of the form

$$\bar{1} \otimes m,$$

and two elements become equal precisely when they differ by something from IM .

Thus the tensor product is measuring the module M after “forgetting” everything killed by the ideal I .

7 One-line intuition for each example

- **Example 1:** reduce each coordinate modulo m .
- **Example 2:** keep only the common torsion shared by $\mathbb{Z}/m$ and $\mathbb{Z}/n$.
- **Example 3:** set $x = 0$ in the module.
- **Example 4:** set $x = 0$ even in a module where $x \neq 0$ but $x^2 = 0$, so the nilpotent x -part disappears.

8 Final universal picture

The construction

$$(A/I) \otimes_A M$$

means that we are viewing the module M after passing from the ring A to the quotient ring A/I .

Equivalently:

- coefficients are reduced modulo I ,
- every element of I acts as zero,
- all elements of IM vanish,
- the remaining information is exactly the quotient module M/IM .

So the identity

$$(A/I) \otimes_A M \cong M/IM$$

is the precise algebraic statement that tensoring with A/I is the same as reducing the module M modulo the action of the ideal I .

Problem — Localization on Spec, base change, and fibers over primes

Exact statement.

- (a) Let A be a ring, S a multiplicatively closed subset of A , and $\varphi : A \rightarrow S^{-1}A$ the canonical homomorphism. Show that the induced map $\varphi^* : \operatorname{Spec}(S^{-1}A) \rightarrow \operatorname{Spec}(A)$ is a homeomorphism onto its image in $X = \operatorname{Spec}(A)$. Write this image as $S^{-1}X$. Show that in the case where $S = \{1, f, f^2, \dots\}$, one has $S^{-1}X = D(f)$.
- (b) Let $\varphi : A \rightarrow B$ be a ring homomorphism, with induced map $\varphi^* : Y = \operatorname{Spec}(B) \rightarrow X = \operatorname{Spec}(A)$. Let $S \subseteq A$ be multiplicatively closed. Show that $\varphi(S)$ is a multiplicatively closed subset of B and that there is an induced morphism $S^{-1}A \rightarrow \varphi(S)^{-1}B$. Thus obtain an induced map $\operatorname{Spec}(\varphi(S)^{-1}B) \rightarrow \operatorname{Spec}(S^{-1}A)$. Show that this map agrees with the restriction of φ^* to $S^{-1}Y$ under the identification of $S^{-1}X, S^{-1}Y$ with $\operatorname{Spec}(S^{-1}A), \operatorname{Spec}(S^{-1}B)$ from part (a). Show that $(\varphi^*)^{-1}(S^{-1}X) = S^{-1}Y$.
- (c) In the same setup as in (b), let $I \subseteq A$ be an ideal and $J = I^e$ its extension in B . Let $\bar{\varphi} : A/I \rightarrow B/J$ be the induced homomorphism. If $\operatorname{Spec}(A/I)$ is identified with $V(I)$ and $\operatorname{Spec}(B/J)$ is identified with $V(J)$, show that $\bar{\varphi}^*$ agrees with the restriction of φ^* to $V(J)$.
- (d) Let $\mathfrak{p} \subseteq A$ be a prime ideal. Take $S = A \setminus \mathfrak{p}$ in (b) and then reduce modulo $\mathfrak{p}A_{\mathfrak{p}}$ as in (c). Deduce that the fiber $(\varphi^*)^{-1}(\mathfrak{p}) \subseteq Y$ is naturally isomorphic to

$$\operatorname{Spec}(B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}}) = \operatorname{Spec}(k(\mathfrak{p}) \otimes_A B),$$

where $k(\mathfrak{p}) = A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}$ is the residue field of the local ring $A_{\mathfrak{p}}$.

Solution.

(a) Let $\varphi : A \rightarrow S^{-1}A$ be the localization map. The induced map on spectra is

$$\varphi^* : \operatorname{Spec}(S^{-1}A) \rightarrow \operatorname{Spec}(A), \quad \mathfrak{P} \mapsto \varphi^{-1}(\mathfrak{P}).$$

Step 1: description of the image and bijection on points. If $\mathfrak{P} \in \operatorname{Spec}(S^{-1}A)$ and $\mathfrak{p} = \varphi^{-1}(\mathfrak{P})$, then $\mathfrak{p} \cap S = \emptyset$. Indeed, for $s \in S$ its image $\varphi(s)$ is a unit in $S^{-1}A$, hence cannot belong to the prime $\mathfrak{P}$. Thus $s \notin \mathfrak{p}$ for all $s \in S$.

Define

$$S^{-1}X := \{\mathfrak{p} \in \operatorname{Spec}(A) \mid \mathfrak{p} \cap S = \emptyset\}.$$

So $\operatorname{im}(\varphi^*) \subseteq S^{-1}X$.

Conversely, if $\mathfrak{p} \in \operatorname{Spec}(A)$ with $\mathfrak{p} \cap S = \emptyset$, then the extension

$$S^{-1}\mathfrak{p} := \left\{ \frac{a}{s} \in S^{-1}A : a \in \mathfrak{p}, s \in S \right\}$$

is a prime ideal of $S^{-1}A$, and one checks

$$\varphi^{-1}(S^{-1}\mathfrak{p}) = \mathfrak{p}.$$

Moreover, if $\mathfrak{P} \in \operatorname{Spec}(S^{-1}A)$ and $\mathfrak{p} = \varphi^{-1}(\mathfrak{P})$, then

$$S^{-1}\mathfrak{p} = \mathfrak{P}.$$

Hence φ^* induces a bijection

$$\mathrm{Spec}(S^{-1}A) \xrightarrow{\sim} S^{-1}X.$$

Step 2: topology (homeomorphism onto the image). A basis of opens in $\mathrm{Spec}(S^{-1}A)$ is given by the principal opens

$$D\left(\frac{a}{s}\right) = \{\mathfrak{P} \in \mathrm{Spec}(S^{-1}A) : a/s \notin \mathfrak{P}\}.$$

We compute the image under φ^* :

$$\varphi^*\left(D\left(\frac{a}{s}\right)\right) = \{\mathfrak{p} \in S^{-1}X : a \notin \mathfrak{p}\} = D(a) \cap S^{-1}X.$$

Thus φ^* identifies basic opens with intersections of basic opens in $\mathrm{Spec}(A)$ with $S^{-1}X$, which is exactly the subspace topology. Therefore φ^* is a homeomorphism $\mathrm{Spec}(S^{-1}A) \cong S^{-1}X$.

Step 3: the principal case $S = \{1, f, f^2, \dots\}$. Then $\mathfrak{p} \cap S = \emptyset$ if and only if $f \notin \mathfrak{p}$. Hence

$$S^{-1}X = \{\mathfrak{p} \in \mathrm{Spec}(A) : f \notin \mathfrak{p}\} = D(f).$$

(b) Let $\varphi : A \rightarrow B$ be a ring homomorphism and $S \subseteq A$ multiplicatively closed. Then $\varphi(S)$ is multiplicatively closed in B since $\varphi(1) = 1$ and $\varphi(st) = \varphi(s)\varphi(t)$.

Define the natural map

$$\theta : S^{-1}A \longrightarrow \varphi(S)^{-1}B, \quad \theta\left(\frac{a}{s}\right) = \frac{\varphi(a)}{\varphi(s)}.$$

This is well-defined and a ring homomorphism (by the universal property of localization). It induces

$$\theta^* : \mathrm{Spec}(\varphi(S)^{-1}B) \rightarrow \mathrm{Spec}(S^{-1}A).$$

By part (a),

$$\mathrm{Spec}(S^{-1}A) \simeq S^{-1}X = \{\mathfrak{p} \in \mathrm{Spec}(A) : \mathfrak{p} \cap S = \emptyset\},$$

and similarly,

$$\mathrm{Spec}(\varphi(S)^{-1}B) \simeq S^{-1}Y = \{\mathfrak{q} \in \mathrm{Spec}(B) : \mathfrak{q} \cap \varphi(S) = \emptyset\}.$$

Now for $\mathfrak{q} \in \mathrm{Spec}(B)$ and $\mathfrak{p} = \varphi^{-1}(\mathfrak{q})$ we have the equivalence

$$\mathfrak{p} \cap S = \emptyset \iff \mathfrak{q} \cap \varphi(S) = \emptyset,$$

because $s \in \mathfrak{p}$ iff $\varphi(s) \in \mathfrak{q}$. Therefore

$$(\varphi^*)^{-1}(S^{-1}X) = S^{-1}Y.$$

Under the above identifications, θ^* is exactly the restriction of φ^* to $S^{-1}Y$.

(c) Let $I \subseteq A$ and let $J = I^e$ be the extension of I in B (i.e. the ideal generated by $\varphi(I)$). The induced map $\overline{\varphi} : A/I \rightarrow B/J$ gives

$$\overline{\varphi}^* : \mathrm{Spec}(B/J) \rightarrow \mathrm{Spec}(A/I).$$

Identify $\operatorname{Spec}(A/I) \simeq V(I) \subseteq \operatorname{Spec}(A)$ via $\mathfrak{p}/I \leftrightarrow \mathfrak{p}$ with $I \subseteq \mathfrak{p}$, and similarly $\operatorname{Spec}(B/J) \simeq V(J) \subseteq \operatorname{Spec}(B)$.

Take $\mathfrak{q} \in V(J)$. Then

$$\overline{\varphi}^*(\mathfrak{q}/J) = \overline{\varphi}^{-1}(\mathfrak{q}/J) = \varphi^{-1}(\mathfrak{q})/I.$$

Under the identifications, this is exactly $\mathfrak{q} \mapsto \varphi^{-1}(\mathfrak{q})$, i.e. $\overline{\varphi}^*$ agrees with the restriction of φ^* to $V(J)$.

(d) Fix $\mathfrak{p} \in \operatorname{Spec}(A)$ and take $S = A \setminus \mathfrak{p}$. Then $S^{-1}A = A_{\mathfrak{p}}$ and $S^{-1}B = B_{\mathfrak{p}}$.

A prime $\mathfrak{q} \in \operatorname{Spec}(B)$ satisfies $\varphi^{-1}(\mathfrak{q}) = \mathfrak{p}$ if and only if its localization $\mathfrak{Q} := \mathfrak{q}B_{\mathfrak{p}}$ satisfies $\mathfrak{Q} \in \operatorname{Spec}(B_{\mathfrak{p}})$ and contracts to the maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$ of $A_{\mathfrak{p}}$. Since $A_{\mathfrak{p}}$ is local with maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$, this is equivalent to $\mathfrak{p}B_{\mathfrak{p}} \subseteq \mathfrak{Q}$. Hence

$$(\varphi^*)^{-1}(\mathfrak{p}) \cong \{\mathfrak{Q} \in \operatorname{Spec}(B_{\mathfrak{p}}) : \mathfrak{p}B_{\mathfrak{p}} \subseteq \mathfrak{Q}\} = \operatorname{Spec}(B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}}).$$

Finally, letting $k(\mathfrak{p}) = A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}$, we have

$$B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}} \cong k(\mathfrak{p}) \otimes_{A_{\mathfrak{p}}} B_{\mathfrak{p}} \cong k(\mathfrak{p}) \otimes_A B,$$

so

$$(\varphi^*)^{-1}(\mathfrak{p}) \cong \operatorname{Spec}(B_{\mathfrak{p}}/\mathfrak{p}B_{\mathfrak{p}}) = \operatorname{Spec}(k(\mathfrak{p}) \otimes_A B).$$

Problem — Geometric interpretation of primary decomposition (irreducible components)

Exact statement. Let X be a topological space. A non-empty subset $Z \subseteq X$ is *irreducible* if whenever $Z = Z_1 \cup Z_2$ with Z_1, Z_2 closed subsets of Z , we have either $Z = Z_1$ or $Z = Z_2$. A topological space is *Noetherian* if it satisfies the descending chain condition for closed subsets.

1. Show that if $Z \subseteq X$ is a closed subset with X a Noetherian topological space, then there is a decomposition

$$Z = Z_1 \cup \cdots \cup Z_r$$

with the Z_i irreducible closed subsets. Show this decomposition is unique up to reordering if it is irredundant, i.e. if we do not have $Z_i \subseteq Z_j$ for any $i \neq j$.

2. Now let A be a ring and $I \subseteq A$ an ideal. Show that I prime implies that $V(I) \subseteq \operatorname{Spec}(A)$ is irreducible.
3. Show that if A is Noetherian, then $\operatorname{Spec}(A)$ is a Noetherian topological space.
4. Show that if A is Noetherian, $Z = V(I)$ and $Z = Z_1 \cup \cdots \cup Z_r$ is an irredundant irreducible decomposition, then there is a one-to-one correspondence between the set $\{Z_i\}$ and the minimal associated primes of A/I .

Solution.

(1) **Existence and uniqueness of irreducible decomposition in a Noetherian space.**

Existence. Assume X is Noetherian and let $Z \subseteq X$ be closed. If Z is irreducible, we are done. Otherwise Z is reducible, so we can write

$$Z = Z' \cup Z'',$$

with Z', Z'' proper closed subsets of Z (hence closed in X). If either Z' or Z'' is reducible, we decompose it further.

We claim this process terminates after finitely many steps. Indeed, in a Noetherian space there is no infinite strictly descending chain of closed subsets. Each time we replace a reducible closed set by a union of two proper closed subsets, we strictly decrease. Therefore the recursion stops, and we obtain a finite decomposition

$$Z = Z_1 \cup \cdots \cup Z_r$$

with each Z_i irreducible and closed.

Uniqueness for irredundant decompositions. Assume we have two irredundant decompositions into irreducible closed subsets:

$$Z = \bigcup_{i=1}^r Z_i \quad \text{and} \quad Z = \bigcup_{j=1}^s W_j,$$

where all Z_i and W_j are irreducible and closed, and no $Z_i \subseteq Z_{i'}$ for $i \neq i'$ (and similarly among the W_j).

Fix i . Since $Z_i \subseteq \bigcup_{j=1}^s W_j$, we have

$$Z_i = \bigcup_{j=1}^s (Z_i \cap W_j),$$

where each $Z_i \cap W_j$ is closed in Z_i . Because Z_i is irreducible, it cannot be covered by finitely many proper closed subsets of itself. Hence there exists some j such that

$$Z_i = Z_i \cap W_j \implies Z_i \subseteq W_j.$$

Now apply the same argument with roles swapped: from $W_j \subseteq \bigcup_{k=1}^r Z_k$ and irreducibility of W_j , there exists k with $W_j \subseteq Z_k$. Thus

$$Z_i \subseteq W_j \subseteq Z_k.$$

By irredundancy of the Z -decomposition this forces $k = i$, hence $Z_i = W_j$. Therefore each Z_i equals some W_j . By symmetry each W_j equals some Z_i . Hence $r = s$ and the two decompositions agree up to reordering.

(2) If I is prime then $V(I)$ is irreducible.

Assume I is prime. Suppose

$$V(I) = Y \cup Z$$

with Y, Z closed subsets of $V(I)$. Since closed subsets in $\text{Spec}(A)$ are of the form $V(J)$, there exist ideals $J, K \subseteq A$ with

$$Y = V(I) \cap V(J) = V(I + J), \quad Z = V(I) \cap V(K) = V(I + K).$$

Then

$$V(I) = V(I + J) \cup V(I + K) = V((I + J) \cap (I + K)) = V((I + J)(I + K)),$$

using $V(J_1) \cup V(J_2) = V(J_1 \cap J_2) = V(J_1 J_2)$.

Equality of closed sets implies equality of radicals:

$$\sqrt{I} = \sqrt{(I + J)(I + K)}.$$

Since I is prime, it is radical, so

$$I = \sqrt{(I + J)(I + K)}.$$

In particular, $(I + J)(I + K) \subseteq I$. Since I is prime, this forces

$$I + J \subseteq I \quad \text{or} \quad I + K \subseteq I,$$

hence $J \subseteq I$ or $K \subseteq I$, which gives

$$V(I + J) = V(I) \quad \text{or} \quad V(I + K) = V(I).$$

Thus $Y = V(I)$ or $Z = V(I)$, proving $V(I)$ is irreducible.

(3) If A is Noetherian then $\text{Spec}(A)$ is Noetherian.

Let

$$V(I_1) \supseteq V(I_2) \supseteq V(I_3) \supseteq \cdots$$

be a descending chain of closed subsets of $\text{Spec}(A)$. Then radicals satisfy the ascending chain

$$\sqrt{I_1} \subseteq \sqrt{I_2} \subseteq \sqrt{I_3} \subseteq \cdots,$$

because $V(I) \subseteq V(J) \iff \sqrt{J} \subseteq \sqrt{I}$. Since A is Noetherian, every ascending chain of ideals stabilizes, hence the chain of radicals stabilizes, and therefore the chain of closed sets stabilizes. Thus $\text{Spec}(A)$ is Noetherian.

(4) Irreducible components of $V(I)$ and minimal associated primes of A/I .

Assume A is Noetherian and let $Z = V(I)$. By (1), Z has an irredundant decomposition into irreducible closed subsets:

$$V(I) = Z_1 \cup \cdots \cup Z_r.$$

Step 1: each irreducible closed subset of $\text{Spec}(A)$ is $V(\mathfrak{p})$ for a unique prime $\mathfrak{p}$. In a Noetherian topological space, every irreducible closed subset has a unique generic point. In $\text{Spec}(A)$, the closure of a point $\mathfrak{p}$ is $V(\mathfrak{p})$. Hence for each i there exists a unique prime $\mathfrak{p}_i$ such that

$$Z_i = \overline{\{\mathfrak{p}_i\}} = V(\mathfrak{p}_i).$$

So we can write

$$V(I) = V(\mathfrak{p}_1) \cup \cdots \cup V(\mathfrak{p}_r).$$

Since $Z_i \subseteq V(I)$, we have $I \subseteq \mathfrak{p}_i$ for each i .

Step 2: irredundancy forces $\mathfrak{p}_i$ to be minimal over I . If $\mathfrak{p}_i$ were not minimal among primes containing I , then there exists a prime $\mathfrak{q}$ with

$$I \subseteq \mathfrak{q} \subsetneq \mathfrak{p}_i.$$

Then $V(\mathfrak{p}_i) \subsetneq V(\mathfrak{q}) \subseteq V(I)$, contradicting that $Z_i = V(\mathfrak{p}_i)$ is an irreducible *component* of $V(I)$ in an irredundant decomposition. Hence each $\mathfrak{p}_i$ is a minimal prime over I .

Conversely, if $\mathfrak{p}$ is minimal over I , then $V(\mathfrak{p})$ is an irreducible closed subset of $V(I)$ and is maximal (by minimality of $\mathfrak{p}$), hence appears as one of the components. Therefore the set $\{Z_i\}$ is in bijection with the set of minimal primes over I .

Step 3: minimal primes over I are the minimal associated primes of A/I (Noetherian case). In a Noetherian ring, the minimal primes over I coincide with the minimal elements of $\text{Ass}(A/I)$. Thus we obtain a one-to-one correspondence

$$\{Z_1, \dots, Z_r\} \longleftrightarrow \{\text{minimal primes over } I\} = \{\text{minimal associated primes of } A/I\},$$

given explicitly by $Z_i = V(\mathfrak{p}_i) \longleftrightarrow \mathfrak{p}_i$.

Problem 2 — Associated primes and a primary decomposition of a monomial ideal

Exact statement. Let

$$I = (x^2, xy, xz, yz) \subseteq k[x, y, z].$$

Find the associated primes of $k[x, y, z]/I$ and a primary decomposition for I .

Solution. Let $R = k[x, y, z]$.

We first determine the minimal primes over I . If a prime ideal $\mathfrak{p}$ contains I , then $x^2 \in \mathfrak{p}$, hence $x \in \mathfrak{p}$. Also $yz \in \mathfrak{p}$, so since $\mathfrak{p}$ is prime, either $y \in \mathfrak{p}$ or $z \in \mathfrak{p}$.

Therefore the minimal primes over I are

$$(x, y) \quad \text{and} \quad (x, z).$$

Now we check for an embedded associated prime. In R/I , consider the class $\bar{x}$ of x . Its annihilator is

$$\text{Ann}_{R/I}(\bar{x}) = \{f \in R : fx \in I\}.$$

Since $x^2, xy, xz \in I$, every element of (x, y, z) annihilates $\bar{x}$. Conversely, if $fx \in I$, then every term of fx must be divisible by one of

$$x^2, xy, xz, yz.$$

Because of the factor x , this forces $f \in (x, y, z)$. Hence

$$\text{Ann}(\bar{x}) = (x, y, z),$$

so (x, y, z) is also an associated prime.

Thus

$$\text{Ass}(R/I) = \{(x, y), (x, z), (x, y, z)\}.$$

Now for a primary decomposition. First,

$$(x, y) \cap (x, z) = (x, yz).$$

Then

$$(x, yz) \cap (x^2, y, z)$$

is the monomial ideal generated by the least common multiples of generators, namely

$$x^2, \quad xy, \quad xz, \quad yz.$$

Therefore

$$I = (x, y) \cap (x, z) \cap (x^2, y, z).$$

This is a primary decomposition because:

- (x, y) and (x, z) are prime ideals, hence primary;
- (x^2, y, z) has radical (x, y, z) , and

$$R/(x^2, y, z) \cong k[x]/(x^2),$$

so it is (x, y, z) -primary.

Hence an irredundant primary decomposition is

$$I = (x, y) \cap (x, z) \cap (x^2, y, z)$$

and

$$\text{Ass}(R/I) = \{(x, y), (x, z), (x, y, z)\}.$$

Problem 3 — Basic properties of support

Exact statement. Show the following statements about the support of an A -module:

- (a) If $0 \rightarrow M_1 \rightarrow M_2 \rightarrow M_3 \rightarrow 0$ is exact, then

$$\text{Supp}(M_2) = \text{Supp}(M_1) \cup \text{Supp}(M_3).$$

- (b) If $M = \sum_i M_i$ for submodules $M_i \subseteq M$, then

$$\text{Supp}(M) = \bigcup_i \text{Supp}(M_i).$$

- (c) For $\mathfrak{p} \subseteq A$ a prime ideal, write

$$k(\mathfrak{p}) := A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}.$$

If M is finitely generated, show that

$$\mathfrak{p} \in \text{Supp}(M) \iff M \otimes_A k(\mathfrak{p}) \neq 0.$$

- (d) If M, N are finitely generated A -modules, then

$$\text{Supp}(M \otimes_A N) = \text{Supp}(M) \cap \text{Supp}(N).$$

- (e) If $f : A \rightarrow B$ is a ring homomorphism and M is a finitely generated A -module, then

$$\text{Supp}(B \otimes_A M) = (f^*)^{-1}(\text{Supp}(M)).$$

[Note: You may find useful the identity on tensor products

$$(M \otimes_A B) \otimes_B C \cong M \otimes_A C$$

given an A -module M , an A -algebra B and a B -algebra C .]

Solution. Recall

$$\text{Supp}(M) = \{\mathfrak{p} \in \text{Spec } A : M_{\mathfrak{p}} \neq 0\}.$$

(a) Localizing the exact sequence at $\mathfrak{p}$ gives

$$0 \rightarrow (M_1)_{\mathfrak{p}} \rightarrow (M_2)_{\mathfrak{p}} \rightarrow (M_3)_{\mathfrak{p}} \rightarrow 0.$$

Hence $(M_2)_{\mathfrak{p}} = 0$ if and only if both $(M_1)_{\mathfrak{p}} = 0$ and $(M_3)_{\mathfrak{p}} = 0$. Therefore

$$\boxed{\text{Supp}(M_2) = \text{Supp}(M_1) \cup \text{Supp}(M_3).}$$

(b) The natural map

$$\bigoplus_i M_i \longrightarrow M$$

is surjective. Localizing at $\mathfrak{p}$, we obtain a surjection

$$\bigoplus_i (M_i)_{\mathfrak{p}} \longrightarrow M_{\mathfrak{p}}.$$

Thus $M_{\mathfrak{p}} \neq 0$ if and only if $(M_i)_{\mathfrak{p}} \neq 0$ for at least one i . Hence

$$\boxed{\text{Supp}(M) = \bigcup_i \text{Supp}(M_i).}$$

(c) Let $N = M_{\mathfrak{p}}$, a finitely generated $A_{\mathfrak{p}}$ -module. Then

$$M \otimes_A k(\mathfrak{p}) \cong M_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} k(\mathfrak{p}) \cong N/\mathfrak{p}A_{\mathfrak{p}}N.$$

By Nakayama's lemma,

$$N = 0 \iff N/\mathfrak{p}A_{\mathfrak{p}}N = 0.$$

Therefore

$$M_{\mathfrak{p}} \neq 0 \iff M \otimes_A k(\mathfrak{p}) \neq 0,$$

so

$$\boxed{\mathfrak{p} \in \text{Supp}(M) \iff M \otimes_A k(\mathfrak{p}) \neq 0.}$$

(d) Localizing at $\mathfrak{p}$,

$$(M \otimes_A N)_{\mathfrak{p}} \cong M_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} N_{\mathfrak{p}}.$$

If $M_{\mathfrak{p}} = 0$ or $N_{\mathfrak{p}} = 0$, then the tensor product is zero.

Conversely, assume $M_{\mathfrak{p}} \neq 0$ and $N_{\mathfrak{p}} \neq 0$. By part (c),

$$M_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} k(\mathfrak{p}) \neq 0, \quad N_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} k(\mathfrak{p}) \neq 0.$$

Hence

$$(M_{\mathfrak{p}} \otimes N_{\mathfrak{p}}) \otimes k(\mathfrak{p}) \cong (M_{\mathfrak{p}} \otimes k(\mathfrak{p})) \otimes_{k(\mathfrak{p})} (N_{\mathfrak{p}} \otimes k(\mathfrak{p})),$$

which is nonzero because it is the tensor product of two nonzero vector spaces over the field $k(\mathfrak{p})$. Thus

$$M_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} N_{\mathfrak{p}} \neq 0.$$

Therefore

$$\boxed{\text{Supp}(M \otimes_A N) = \text{Supp}(M) \cap \text{Supp}(N).}$$

(e) Let $\mathfrak{q} \in \text{Spec } B$, and set $\mathfrak{p} = f^{-1}(\mathfrak{q}) \in \text{Spec } A$. By part (c),

$$\mathfrak{q} \in \text{Supp}_B(B \otimes_A M) \iff (B \otimes_A M) \otimes_B k(\mathfrak{q}) \neq 0.$$

Using the tensor identity,

$$(B \otimes_A M) \otimes_B k(\mathfrak{q}) \cong M \otimes_A k(\mathfrak{q}).$$

Since $k(\mathfrak{q})$ is naturally a $k(\mathfrak{p})$ -algebra,

$$M \otimes_A k(\mathfrak{q}) \cong (M \otimes_A k(\mathfrak{p})) \otimes_{k(\mathfrak{p})} k(\mathfrak{q}).$$

This is nonzero if and only if $M \otimes_A k(\mathfrak{p}) \neq 0$. By part (c), this is equivalent to $\mathfrak{p} \in \text{Supp}_A(M)$. Hence

$$\mathfrak{q} \in \text{Supp}_B(B \otimes_A M) \iff f^{-1}(\mathfrak{q}) \in \text{Supp}_A(M),$$

which proves

$$\boxed{\text{Supp}(B \otimes_A M) = (f^*)^{-1}(\text{Supp}(M)).}$$

Problem 4 — Extending ideals to a polynomial ring

Exact statement. Let A be a ring and let $A[x]$ be the polynomial ring. If $I \subseteq A$ is an ideal, denote by $I[x]$ the set of all polynomials in $A[x]$ with coefficients in I .

- (a) Show $I[x]$ is the extension I^e of I to $A[x]$.
- (b) If $\mathfrak{p}$ is a prime ideal of A , show that $\mathfrak{p}[x]$ is a prime ideal of $A[x]$.
- (c) If $\mathfrak{q}$ is a $\mathfrak{p}$ -primary ideal of A , then $\mathfrak{q}[x]$ is a $\mathfrak{p}[x]$ -primary ideal in $A[x]$.
- (d) If

$$I = \bigcap_{i=1}^n \mathfrak{q}_i$$

is an irredundant primary decomposition of I in A , then

$$I[x] = \bigcap_{i=1}^n \mathfrak{q}_i[x]$$

is an irredundant primary decomposition in $A[x]$.

- (e) If $\mathfrak{p}$ is a minimal prime of $V(I)$, then $\mathfrak{p}[x]$ is a minimal prime of $V(I[x])$.

Solution.

(a) The extension of I to $A[x]$ is

$$I^e = IA[x].$$

A polynomial belongs to $IA[x]$ if and only if all its coefficients lie in I . Hence

$$\boxed{I[x] = IA[x] = I^e.}$$

(b) We have

$$A[x]/\mathfrak{p}[x] \cong (A/\mathfrak{p})[x].$$

Since $\mathfrak{p}$ is prime, $A/\mathfrak{p}$ is an integral domain, so $(A/\mathfrak{p})[x]$ is an integral domain. Therefore

$$\boxed{\mathfrak{p}[x] \text{ is a prime ideal of } A[x].}$$

(c) Assume $\mathfrak{q}$ is $\mathfrak{p}$ -primary in A . Then

$$A[x]/\mathfrak{q}[x] \cong (A/\mathfrak{q})[x].$$

Also,

$$\sqrt{\mathfrak{q}[x]} = \sqrt{\mathfrak{q}}[x] = \mathfrak{p}[x].$$

So it remains to show that every zero-divisor in $(A/\mathfrak{q})[x]$ is nilpotent.

Since $\mathfrak{q}$ is primary, every zero-divisor in $A/\mathfrak{q}$ is nilpotent. If $f(x) \in (A/\mathfrak{q})[x]$ is a zero-divisor, McCoy's theorem gives a nonzero element $a \in A/\mathfrak{q}$ such that

$$af(x) = 0.$$

Therefore every coefficient of $f(x)$ is annihilated by a , hence is a zero-divisor in $A/\mathfrak{q}$, and thus nilpotent. A polynomial with finitely many nilpotent coefficients is nilpotent. Hence every zero-divisor in $(A/\mathfrak{q})[x]$ is nilpotent, so

$$\boxed{\mathfrak{q}[x] \text{ is } \mathfrak{p}[x]\text{-primary.}}$$

(d) Since coefficients are checked componentwise,

$$\left(\bigcap_{i=1}^n \mathfrak{q}_i \right) [x] = \bigcap_{i=1}^n \mathfrak{q}_i [x].$$

Thus

$$I[x] = \bigcap_{i=1}^n \mathfrak{q}_i [x].$$

By part (c), each $\mathfrak{q}_i[x]$ is primary, so this is a primary decomposition.

To prove irredundancy, suppose $\mathfrak{q}_j[x]$ were redundant. Then

$$\bigcap_{i \neq j} \mathfrak{q}_i [x] \subseteq \mathfrak{q}_j [x].$$

But because the original decomposition is irredundant, there exists

$$a \in \bigcap_{i \neq j} \mathfrak{q}_i \quad \text{with} \quad a \notin \mathfrak{q}_j.$$

Viewing a as a constant polynomial gives

$$a \in \bigcap_{i \neq j} \mathfrak{q}_i[x] \quad \text{but} \quad a \notin \mathfrak{q}_j[x],$$

a contradiction. Hence

$$I[x] = \bigcap_{i=1}^n \mathfrak{q}_i[x]$$

is an irredundant primary decomposition in $A[x]$.

(e) Assume $\mathfrak{p}$ is minimal among primes containing I . By part (b), $\mathfrak{p}[x]$ is prime and clearly contains $I[x]$, so $\mathfrak{p}[x] \in V(I[x])$.

Now let P be a prime ideal of $A[x]$ with

$$I[x] \subseteq P \subseteq \mathfrak{p}[x].$$

Contracting to A , we get

$$I \subseteq P \cap A \subseteq \mathfrak{p}.$$

By minimality of $\mathfrak{p}$ over I , it follows that

$$P \cap A = \mathfrak{p}.$$

Hence every element of $\mathfrak{p}$ lies in P , so all constant polynomials from $\mathfrak{p}$ lie in P , and therefore

$$\mathfrak{p}[x] \subseteq P.$$

Since also $P \subseteq \mathfrak{p}[x]$, we obtain $P = \mathfrak{p}[x]$.

Thus

$$\boxed{\mathfrak{p}[x] \text{ is a minimal prime of } V(I[x]).}$$

Theory for Problems 2–4

1. Associated primes and primary decomposition

Associated primes. Let R be a ring and M an R -module. A prime ideal $\mathfrak{p} \subseteq R$ is called an associated prime of M if

$$\mathfrak{p} = \text{Ann}(m) = \{r \in R : rm = 0\}$$

for some $m \in M$. The set of associated primes is denoted

$$\text{Ass}(M).$$

When $M = R/I$, one has

$$\text{Ann}_{R/I}(\bar{f}) = (I : f),$$

so

$$\mathfrak{p} \in \text{Ass}(R/I) \iff \mathfrak{p} = (I : f) \text{ for some } f \in R.$$

Minimal and embedded associated primes. Among the associated primes of R/I , some are minimal primes over I . Others may properly contain minimal ones; these are called embedded primes.

Primary ideals. An ideal $Q \subseteq R$ is primary if whenever

$$ab \in Q, \quad a \notin Q,$$

then $b^n \in Q$ for some $n \geq 1$.

Equivalently, in R/Q , every zero-divisor is nilpotent.

If Q is primary, then $\sqrt{Q}$ is prime. If $\sqrt{Q} = \mathfrak{p}$, then Q is called $\mathfrak{p}$ -primary.

Primary decomposition. A primary decomposition of I is an expression

$$I = Q_1 \cap \cdots \cap Q_n$$

with each Q_i primary. If no Q_i can be omitted, the decomposition is irredundant.

In a Noetherian ring, every ideal admits a primary decomposition.

Monomial ideals. For monomial ideals in $k[x_1, \dots, x_n]$, intersections remain monomial, and they can be computed using least common multiples of generators. In particular, the intersection of two monomial ideals is generated by all

$$\text{lcm}(a_i, b_j)$$

where a_i and b_j run through the monomial generators of the two ideals.

Colon ideals and annihilators. For $I \subseteq R$ and $f \in R$, define

$$(I : f) = \{r \in R : rf \in I\}.$$

If $M = R/I$, then

$$\text{Ann}(\bar{f}) = (I : f).$$

This is one of the standard tools for finding associated primes of R/I .

2. Support of a module

Definition. For an A -module M , the support is

$$\text{Supp}(M) = \{\mathfrak{p} \in \text{Spec}(A) : M_{\mathfrak{p}} \neq 0\}.$$

Localization viewpoint. The localization $M_{\mathfrak{p}}$ measures the behavior of M near the prime $\mathfrak{p}$. Thus $\mathfrak{p} \in \text{Supp}(M)$ exactly when M does not vanish near $\mathfrak{p}$.

Support and exact sequences. Since localization is exact, from

$$0 \rightarrow M_1 \rightarrow M_2 \rightarrow M_3 \rightarrow 0$$

one gets

$$0 \rightarrow (M_1)_{\mathfrak{p}} \rightarrow (M_2)_{\mathfrak{p}} \rightarrow (M_3)_{\mathfrak{p}} \rightarrow 0.$$

Hence

$$\text{Supp}(M_2) = \text{Supp}(M_1) \cup \text{Supp}(M_3).$$

Residue fields and pointwise testing. For a prime $\mathfrak{p}$, define

$$k(\mathfrak{p}) = A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}.$$

If M is finitely generated, then

$$\mathfrak{p} \in \text{Supp}(M) \iff M \otimes_A k(\mathfrak{p}) \neq 0.$$

This follows from

$$M \otimes_A k(\mathfrak{p}) \cong M_{\mathfrak{p}}/\mathfrak{p}M_{\mathfrak{p}}$$

and Nakayama's lemma.

Support of tensor products. For finitely generated A -modules M, N ,

$$\text{Supp}(M \otimes_A N) = \text{Supp}(M) \cap \text{Supp}(N).$$

This comes from localization and the fact that the tensor product of two nonzero vector spaces over a field is nonzero.

Base change. If $f : A \rightarrow B$ is a ring homomorphism and M is a finitely generated A -module, then

$$\text{Supp}_B(B \otimes_A M) = (f^*)^{-1}(\text{Supp}_A(M)),$$

where

$$f^* : \text{Spec}(B) \rightarrow \text{Spec}(A), \quad \mathfrak{q} \mapsto f^{-1}(\mathfrak{q}).$$

Support versus associated primes. For finitely generated modules over a Noetherian ring,

$$\text{Supp}(M) = V(\text{Ann}(M))$$

and also

$$\text{Supp}(M) = \bigcup_{\mathfrak{p} \in \text{Ass}(M)} V(\mathfrak{p}).$$

3. Extension of ideals to polynomial rings

Extension and contraction. If $A \subseteq B$ is a ring extension and $I \subseteq A$ is an ideal, the extension of I to B is

$$I^e = IB.$$

For the inclusion $A \hookrightarrow A[x]$, this becomes

$$I^e = IA[x].$$

But this is exactly the set of polynomials all of whose coefficients lie in I , so

$$I[x] = IA[x].$$

Quotients of polynomial rings. A basic identity is

$$A[x]/I[x] \cong (A/I)[x].$$

This reduces many questions about $I[x]$ to the quotient ring A/I .

Prime ideals extend to prime ideals. If $\mathfrak{p} \subseteq A$ is prime, then

$$A[x]/\mathfrak{p}[x] \cong (A/\mathfrak{p})[x].$$

Since $A/\mathfrak{p}$ is a domain, $(A/\mathfrak{p})[x]$ is also a domain. Hence $\mathfrak{p}[x]$ is prime.

Primary ideals extend to primary ideals. If $\mathfrak{q}$ is $\mathfrak{p}$ -primary in A , then $\mathfrak{q}[x]$ is $\mathfrak{p}[x]$ -primary in $A[x]$. Indeed,

$$\sqrt{\mathfrak{q}[x]} = \sqrt{\mathfrak{q}}[x] = \mathfrak{p}[x],$$

and

$$A[x]/\mathfrak{q}[x] \cong (A/\mathfrak{q})[x].$$

The zero-divisors in $(A/\mathfrak{q})[x]$ are nilpotent, so the quotient is primary.

Primary decompositions extend coefficientwise. If

$$I = \bigcap_{i=1}^n Q_i,$$

then

$$I[x] = \bigcap_{i=1}^n Q_i[x].$$

This is because a polynomial belongs to $Q_i[x]$ exactly when each coefficient lies in Q_i .

Irredundancy is preserved. If

$$I = \bigcap_{i=1}^n Q_i$$

is irredundant, then

$$I[x] = \bigcap_{i=1}^n Q_i[x]$$

is also irredundant. Otherwise one would get a contradiction by looking at constant polynomials.

Minimal primes. If $\mathfrak{p}$ is a minimal prime over I , then $\mathfrak{p}[x]$ is a minimal prime over $I[x]$.

4. Key lemmas behind these problems

Nakayama's lemma. If $(R, \mathfrak{m})$ is local and M is a finitely generated R -module, then

$$M = \mathfrak{m}M \implies M = 0.$$

Equivalently,

$$M = 0 \iff M/\mathfrak{m}M = 0.$$

Exactness of localization. Localization preserves exact sequences:

$$0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0 \implies 0 \rightarrow M'_\mathfrak{p} \rightarrow M_\mathfrak{p} \rightarrow M''_\mathfrak{p} \rightarrow 0.$$

Tensor product and localization. One has

$$(M \otimes_A N)_\mathfrak{p} \cong M_\mathfrak{p} \otimes_{A_\mathfrak{p}} N_\mathfrak{p}.$$

Tensor-associativity under base change. If B is an A -algebra and C a B -algebra, then

$$(M \otimes_A B) \otimes_B C \cong M \otimes_A C.$$

5. Quick intuition summary

Problem 2. The main ideas are:

- minimal primes over a monomial ideal,
- annihilators $(I : f)$,
- monomial intersections via least common multiples,
- primary components and their radicals.

Problem 3. The main ideas are:

- support as the set where localization is nonzero,
- exactness of localization,
- residue fields $k(\mathfrak{p})$,
- Nakayama's lemma,
- tensor product as intersection of supports,
- base change as inverse image of support.

Problem 4. The main ideas are:

- extension $I \mapsto I[x]$,
- the quotient identity

$$A[x]/I[x] \cong (A/I)[x],$$

- preservation of prime and primary ideals,
- preservation of irredundant primary decompositions,
- extension of minimal primes.

Good examples of the geometric meaning of ideals

Assume k is a field.

1. The zero ideal. In $k[x, y]$, let

$$I = (0).$$

Then

$$V(I) = V(0) = k^2,$$

since the zero polynomial vanishes everywhere.

2. The unit ideal. Let

$$I = (1).$$

Then

$$V(I) = \emptyset,$$

because 1 does not vanish at any point.

3. A point. Let

$$I = (x - a, y - b) \subseteq k[x, y].$$

Then

$$V(I) = \{(a, b)\}.$$

Moreover,

$$k[x, y]/(x - a, y - b) \cong k.$$

4. A line. Let

$$I = (y - mx - c) \subseteq k[x, y].$$

Then

$$V(I) = \{(x, y) \in k^2 : y = mx + c\}.$$

Also,

$$k[x, y]/(y - mx - c) \cong k[x].$$

5. A parabola. Let

$$I = (y - x^2) \subseteq k[x, y].$$

Then

$$V(I) = \{(x, y) \in k^2 : y = x^2\},$$

and

$$k[x, y]/(y - x^2) \cong k[x].$$

6. A circle. Over $k = \mathbb{R}$, let

$$I = (x^2 + y^2 - 1) \subseteq \mathbb{R}[x, y].$$

Then

$$V(I) = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}.$$

7. Union of the coordinate axes. Let

$$I = (xy) \subseteq k[x, y].$$

Then

$$V(I) = \{(x, y) : xy = 0\} = V(x) \cup V(y).$$

Thus $V(I)$ is the union of the two coordinate axes.

8. Intersections of ideals and unions of varieties. Since

$$(x) \cap (y) = (xy)$$

in $k[x, y]$, we get

$$V((x) \cap (y)) = V(xy) = V(x) \cup V(y).$$

This illustrates the rule

$$V(I \cap J) = V(I) \cup V(J).$$

9. Two points. Let

$$I = (x, y) \cap (x - 1, y) \subseteq k[x, y].$$

Then

$$V(I) = \{(0, 0), (1, 0)\}.$$

10. Same geometric set, different ideals. Let

$$I = (y), \quad J = (y^2) \subseteq k[x, y].$$

Then

$$V(I) = V(J) = \{(x, y) : y = 0\}.$$

However,

$$k[x, y]/(y) \cong k[x],$$

while

$$k[x, y]/(y^2)$$

contains a nonzero nilpotent element $\bar{y}$ with $\bar{y}^2 = 0$. So the ideals define the same set, but different quotient-ring structures.

11. A cusp. Let

$$I = (y^2 - x^3) \subseteq k[x, y].$$

Then

$$V(I) = \{(x, y) : y^2 = x^3\},$$

a singular plane curve.

12. A coordinate plane in k^3 . In $k[x, y, z]$, let

$$I = (z).$$

Then

$$V(I) = \{(x, y, z) \in k^3 : z = 0\},$$

the xy -plane.

13. A line in space. Let

$$I = (x - 1, y + 2) \subseteq k[x, y, z].$$

Then

$$V(I) = \{(1, -2, z) : z \in k\},$$

a line parallel to the z -axis.

14. A finite grid of points. Let

$$I = (x(x - 1), y(y - 1)) \subseteq k[x, y].$$

Then

$$V(I) = \{(0, 0), (0, 1), (1, 0), (1, 1)\}.$$

15. Reversal of inclusions. If

$$I \subseteq J,$$

then

$$V(J) \subseteq V(I).$$

For example,

$$(xy) \subseteq (x)$$

and therefore

$$V(x) \subseteq V(xy).$$

Key rules. The basic geometric rules are

$$V(I \cap J) = V(I) \cup V(J), \quad V(I + J) = V(I) \cap V(J).$$

Thus intersection of ideals corresponds to union of algebraic sets, while sum of ideals corresponds to intersection of algebraic sets.

Deep theory: polynomials, ideals, quotients, annihilators, centers, algebras

Throughout, rings have 1. For ideals, quotient rings, support, prime ideals, and polynomial geometry, the default setting is *commutative rings* unless explicitly stated otherwise. For the notion of *center*, we must allow possibly noncommutative rings.

1. Polynomial rings

1.1. What is a polynomial ring?

If A is a ring, then

$$A[x]$$

is the ring of formal expressions

$$a_0 + a_1x + \cdots + a_nx^n, \quad a_i \in A,$$

with the usual addition and multiplication.

The key word is *formal*. The symbol x is not initially a number; it is an indeterminate. So in $A[x]$, the expression $x^2 + 1$ is an element of a ring, not yet a function.

More generally,

$$A[x_1, \dots, x_n]$$

is the polynomial ring in several indeterminates.

1.2. Why polynomial rings matter

Polynomial rings are the basic way of creating new rings from old ones by adjoining formal variables.

They are the algebraic place where you write equations.

For example:

$$\mathbb{Z}[x]$$

is where integer-coefficient polynomial equations live,

$$k[x]$$

is the one-variable coordinate ring over a field k ,

$$k[x, y]$$

is the ring of polynomial functions in two variables, and

$$k[x, y, z]$$

is the basic ring for affine 3-space.

1.3. Universal property of $A[x]$

This is one of the deepest ideas.

If B is an A -algebra and $b \in B$, then there is a unique A -algebra homomorphism

$$\varphi : A[x] \rightarrow B$$

such that

$$\varphi(x) = b.$$

So $A[x]$ is the “most general” A -algebra generated by one element.

Likewise,

$$A[x_1, \dots, x_n]$$

is the most general A -algebra generated by n commuting elements.

This is why polynomial rings are called *free commutative A -algebras* on generators $x_1, \dots, x_n$.

1.4. Evaluation maps

Given $a \in A$, you can evaluate a polynomial at a :

$$\text{ev}_a : A[x] \rightarrow A, \quad f(x) \mapsto f(a).$$

This is a ring homomorphism.

A fundamental fact is

$$\ker(\text{ev}_a) = (x - a),$$

so

$$A[x]/(x - a) \cong A.$$

This is one of the first and most important quotient examples.

1.5. Deep example: $k[x]/(x - a)$

Take k a field and $a \in k$. Define

$$\varphi : k[x] \rightarrow k, \quad f(x) \mapsto f(a).$$

Then:

- φ is surjective;
- the kernel is $(x - a)$.

So by the first isomorphism theorem,

$$k[x]/(x - a) \cong k.$$

Interpretation: in the quotient, the relation $x = a$ has been imposed.

So quotienting by an ideal means: *force some equations to hold*.

1.6. Multivariable viewpoint

In

$$k[x, y],$$

the ideal $(y - x^2)$ imposes the relation

$$y = x^2.$$

Hence

$$k[x, y]/(y - x^2) \cong k[x].$$

This quotient is the coordinate ring of the parabola $y = x^2$.

So polynomial rings plus quotient rings are the algebraic language of geometry.

2. Ideals**2.1. Definition**

An ideal $I \subseteq A$ is a subset such that:

1. I is an additive subgroup of A ;
2. for all $a \in A$ and $r \in I$,

$$ar \in I.$$

In a commutative ring, this means $AI \subseteq I$.

Ideals are the right objects to quotient by. Subrings are usually not.

2.2. Why ideals are the right kernels

If

$$f : A \rightarrow B$$

is a ring homomorphism, then

$$\ker(f)$$

is always an ideal of A .

Conversely, every ideal $I \subseteq A$ is the kernel of the quotient map

$$A \rightarrow A/I.$$

So ideals are exactly the algebraic objects that describe “relations” or “constraints”.

2.3. Basic operations on ideals

If $I, J \subseteq A$ are ideals, then:

Sum:

$$I + J = \{a + b : a \in I, b \in J\}.$$

Intersection:

$$I \cap J.$$

Product:

$$IJ = \left\{ \sum_k a_k b_k : a_k \in I, b_k \in J \right\}.$$

Colon:

$$(I : J) = \{r \in A : rJ \subseteq I\}.$$

For a single element $f \in A$,

$$(I : f) = \{r \in A : rf \in I\}.$$

This colon ideal is extremely important for annihilators and associated primes.

2.4. Types of ideals

Principal ideal. Generated by one element:

$$(a) = aA.$$

Examples:

$$(6) \subseteq \mathbb{Z}, \quad (x^2 + 1) \subseteq k[x].$$

Finitely generated ideal.

$$(a_1, \dots, a_n) = a_1A + \dots + a_nA.$$

Example:

$$(x^2, xy, y^3) \subseteq k[x, y].$$

Prime ideal. An ideal $\mathfrak{p} \neq A$ is prime if

$$ab \in \mathfrak{p} \implies a \in \mathfrak{p} \text{ or } b \in \mathfrak{p}.$$

Equivalently,

$$A/\mathfrak{p} \text{ is an integral domain.}$$

Maximal ideal. An ideal $\mathfrak{m} \neq A$ is maximal if there is no ideal strictly between $\mathfrak{m}$ and A . Equivalently,

$$A/\mathfrak{m} \text{ is a field.}$$

Radical ideal.

$$\sqrt{I} = \{a \in A : a^n \in I \text{ for some } n \geq 1\}.$$

If $\sqrt{I} = I$, then I is radical.

Equivalently,

$$A/I$$

has no nonzero nilpotents, that is, A/I is reduced.

Primary ideal. An ideal Q is primary if

$$ab \in Q, \quad a \notin Q \implies b^n \in Q \text{ for some } n.$$

Equivalently, all zero-divisors in A/Q are nilpotent.

2.5. Geometric meaning of ideals

In

$$k[x_1, \dots, x_n],$$

an ideal I defines a set

$$V(I) = \{a \in k^n : f(a) = 0 \text{ for all } f \in I\}.$$

So:

- ideals correspond to algebraic sets;
- larger ideals give smaller geometric sets;
- quotient rings encode the functions on those sets.

Example:

$$I = (xy) \subseteq k[x, y].$$

Then

$$V(I) = V(xy) = V(x) \cup V(y),$$

the union of the two coordinate axes.

This already shows that one equation can define a reducible space.

2.6. Deep example: prime vs. radical vs. primary

Take

$$I = (xy) \subseteq k[x, y].$$

Then:

- I is not prime, because $xy \in I$ but $x \notin I$ and $y \notin I$;
- I is radical if xy is squarefree;
- I is not primary.

Indeed,

$$k[x, y]/(xy)$$

has zero-divisors ($\bar{x}\bar{y} = 0$), but $\bar{x}$ and $\bar{y}$ are not nilpotent.

So:

- prime is stronger than radical;
- primary is different from radical;
- quotients reveal these properties.

3. Quotient rings

3.1. Definition

If $I \subseteq A$ is an ideal, then

$$A/I$$

is the set of cosets

$$a + I,$$

with operations

$$(a + I) + (b + I) = (a + b) + I, \quad (a + I)(b + I) = ab + I.$$

In the quotient, every element of I becomes zero.

So quotienting by I means: identify all elements of I with 0.

3.2. Quotients as imposed relations

This is the most useful intuition:

- in $A[x]/(x - a)$, we impose $x = a$;
- in $k[x]/(x^2)$, we impose $x^2 = 0$;
- in $k[x, y]/(y - x^2)$, we impose $y = x^2$;
- in $k[x, y]/(x, y)$, we impose $x = 0$ and $y = 0$.

So quotient rings are rings where chosen equations are declared true.

3.3. First isomorphism theorem

If

$$f : A \rightarrow B$$

is a ring homomorphism, then

$$A/\ker(f) \cong \operatorname{Im}(f).$$

This explains most standard quotient computations.

3.4. Correspondence theorem

The ideals of A/I correspond exactly to the ideals of A containing I .

More precisely:

$$J \subseteq A, \quad I \subseteq J \quad \longleftrightarrow \quad J/I \subseteq A/I.$$

This is one of the most important structural facts.

It tells you that quotienting does not destroy ideal theory; it compresses it.

3.5. Deep example: dual numbers

Consider

$$R = k[\varepsilon]/(\varepsilon^2).$$

Write $\bar{\varepsilon}$ for the class of ε . Then

$$\bar{\varepsilon}^2 = 0, \quad \bar{\varepsilon} \neq 0.$$

So R has a nonzero nilpotent element.

This means:

- R is not a domain;
- R is not reduced;
- the ideal $(\varepsilon^2) \subseteq k[\varepsilon]$ is not radical.

This ring is extremely important in geometry and deformation theory because it encodes an “infinitesimal thickening” of a point.

3.6. Deep example: quotient of a polynomial ring by a curve equation

Take

$$R = k[x, y]/(y - x^2).$$

Define

$$\varphi : k[x, y] \rightarrow k[t], \quad x \mapsto t, \quad y \mapsto t^2.$$

Then φ is surjective and

$$\ker(\varphi) = (y - x^2).$$

Hence

$$k[x, y]/(y - x^2) \cong k[t].$$

So algebraically, the parabola has the same coordinate ring as a polynomial ring in one variable.

Geometric meaning: the parabola is just an affine line in disguise.

4. Annihilators

4.1. Definition for a module

Let M be an A -module. Its annihilator is

$$\text{Ann}_A(M) = \{a \in A : am = 0 \text{ for all } m \in M\}.$$

This is an ideal of A .

It measures how much of the ring acts trivially on the whole module.

4.2. Annihilator of an element

For a single element $m \in M$,

$$\text{Ann}_A(m) = \{a \in A : am = 0\}.$$

This is also an ideal.

The annihilator of an element tells you what kills that particular element.

4.3. First key example

If $M = A/I$, then

$$\text{Ann}_A(A/I) = I.$$

Why? Because a kills every class in A/I exactly when $a \in I$.

So quotient rings are the simplest modules whose annihilator is visibly known.

4.4. Colon ideals and annihilators

If $M = A/I$ and $f \in A$, then the annihilator of the class $\bar{f} \in A/I$ is

$$\text{Ann}_A(\bar{f}) = (I : f).$$

This is a major computational tool.

Indeed,

$$a\bar{f} = 0 \iff af \in I.$$

So colon ideals are exactly annihilators inside quotient modules.

4.5. Deep example: annihilator in dual numbers

Take

$$R = k[\varepsilon]/(\varepsilon^2).$$

Then for the element $\bar{\varepsilon} \in R$,

$$\text{Ann}_R(\bar{\varepsilon}) = (\bar{\varepsilon}).$$

Because

$$(a + b\bar{\varepsilon})\bar{\varepsilon} = a\bar{\varepsilon} + b\bar{\varepsilon}^2 = a\bar{\varepsilon},$$

this is zero if and only if $a = 0$, so the annihilator is exactly the ideal generated by $\bar{\varepsilon}$.

This is a good picture of how annihilators detect nilpotent directions.

4.6. Deep example from an earlier problem

Let

$$I = (x^2, xy, xz, yz) \subseteq k[x, y, z].$$

In R/I , take $\bar{x}$. Then

$$\text{Ann}_R(\bar{x}) = (I : x).$$

Since

$$x \cdot x = x^2 \in I, \quad yx = xy \in I, \quad zx = xz \in I,$$

we get

$$(x, y, z) \subseteq (I : x).$$

Conversely, if $fx \in I$, then every term of f must lie in (x, y, z) . So

$$(I : x) = (x, y, z).$$

Hence

$$\text{Ann}_R(\bar{x}) = (x, y, z),$$

which gives an associated prime.

This is a perfect example of how annihilators and associated primes are linked.

4.7. Why annihilators matter

Annihilators are central in:

- module structure;
- associated primes;
- support;
- torsion theory;
- decomposition theory.

A prime ideal $\mathfrak{p}$ is associated to M when

$$\mathfrak{p} = \text{Ann}(m)$$

for some $m \in M$.

So annihilators are the entry point to associated primes.

5. Centers

5.1. Definition

If R is any ring, possibly noncommutative, its center is

$$Z(R) = \{z \in R : zr = rz \text{ for all } r \in R\}.$$

These are the elements that commute with everything.

The center is a commutative subring of R .

5.2. Why the center matters

The center tells you how far the ring is from being commutative.

If R is commutative, then

$$Z(R) = R.$$

If R is noncommutative, the center is usually much smaller.

It plays the role of the “scalar part” of the ring.

5.3. Example: matrix ring

For a field k ,

$$Z(M_n(k)) = kI_n.$$

So the only matrices that commute with every matrix are the scalar matrices.

This is a fundamental theorem.

Idea of proof: use the matrix units E_{ij} . If A commutes with all E_{ij} , then A must have equal diagonal entries and zero off-diagonal entries.

So the center is

$$\{\lambda I_n : \lambda \in k\}.$$

5.4. Example: commutative polynomial ring

Since

$$k[x_1, \dots, x_n]$$

is commutative,

$$Z(k[x_1, \dots, x_n]) = k[x_1, \dots, x_n].$$

So in ordinary commutative algebra, the center is all of the ring.

That is why “center” becomes interesting mostly when one studies matrix rings, group algebras, operator rings, and other noncommutative objects.

5.5. Example: group algebra

If $k[G]$ is the group algebra of a finite group G , then its center consists of linear combinations of class sums.

So the center reflects conjugacy structure in the group.

This is already a deep bridge between group theory and ring theory.

6. Algebras

6.1. Definition of a k -algebra

A k -algebra B is a ring together with a ring homomorphism

$$k \rightarrow Z(B).$$

So elements of k act as scalars, and they must commute with elements of B .

In the commutative setting, one often just says a k -algebra is a commutative ring B with a ring map

$$k \rightarrow B.$$

The center appears in the general definition because scalars should behave centrally.

6.2. Why algebras matter

A k -algebra is simultaneously:

- a ring;
- a k -vector space or k -module;
- equipped with multiplication compatible with scalar multiplication.

So algebra combines linear structure and ring structure.

6.3. Standard examples

Polynomial algebra.

$$k[x_1, \dots, x_n]$$

is a k -algebra.

Quotient algebra. If

$$I \subseteq k[x_1, \dots, x_n],$$

then

$$k[x_1, \dots, x_n]/I$$

is again a k -algebra.

These are the coordinate rings of affine algebraic sets.

Matrix algebra.

$$M_n(k)$$

is a k -algebra, but usually noncommutative.

Product algebra.

$$k \times k$$

is a commutative k -algebra.

6.4. Free commutative k -algebras

The polynomial ring

$$k[x_1, \dots, x_n]$$

is the free commutative k -algebra on n generators.

This means: to define a k -algebra map from

$$k[x_1, \dots, x_n],$$

it is enough to choose images of the variables.

So if B is a commutative k -algebra and $b_1, \dots, b_n \in B$, there is a unique k -algebra map

$$k[x_1, \dots, x_n] \rightarrow B, \quad x_i \mapsto b_i.$$

This is one of the deepest organizing principles in algebra.

6.5. Finitely generated algebras

A k -algebra B is finitely generated if

$$B \cong k[x_1, \dots, x_n]/I$$

for some ideal I .

These are the algebras that arise as affine coordinate rings.

So affine algebraic geometry studies exactly finitely generated commutative k -algebras.

6.6. Deep geometric example

Let

$$B = k[x, y]/(y^2 - x^3).$$

This is a finitely generated k -algebra. It is the coordinate ring of the cusp

$$y^2 = x^3.$$

So one quotient ring captures a whole singular geometric object.

This is why quotient algebras are so central: they encode spaces by equations.

7. The links between all these notions

Here is the conceptual chain.

7.1. Polynomial rings create spaces of equations

You begin with

$$k[x_1, \dots, x_n].$$

7.2. Ideals collect systems of equations

An ideal

$$I \subseteq k[x_1, \dots, x_n]$$

means “all algebraic consequences of some chosen equations”.

7.3. Quotients impose those equations

The quotient

$$k[x_1, \dots, x_n]/I$$

is the ring where every polynomial in I becomes zero.

7.4. Annihilators detect what acts trivially

Inside a quotient or module, the annihilator tells you what kills an element or the whole module.

7.5. The center identifies scalar-like elements

In noncommutative contexts, the center is the part that still behaves like scalars.

7.6. Algebras organize all of this over a base ring or field

A k -algebra is a ring built over k , and polynomial rings plus quotient rings are the basic examples.

8. Deep examples tying everything together

Example 1. $(A/I)[x] \cong A[x]/I[x]$

This is one of the most useful facts in recent problems.

Define

$$\varphi : A[x] \rightarrow (A/I)[x]$$

by reducing coefficients modulo I . Then:

- φ is surjective;
- the kernel is exactly the polynomials whose coefficients lie in I , namely $I[x]$.

Therefore

$$A[x]/I[x] \cong (A/I)[x].$$

Interpretation: quotient first or adjoin x first — same result.

Example 2. Quotient that becomes a field

Take

$$\mathbb{Z}[x]/(2, x).$$

Since $x = 0$ and $2 = 0$ in the quotient, every class is determined by an integer modulo 2. Hence

$$\mathbb{Z}[x]/(2, x) \cong \mathbb{F}_2.$$

So $(2, x)$ is a maximal ideal of $\mathbb{Z}[x]$.

Example 3. Quotient that becomes nonreduced

Take

$$k[x]/(x^2).$$

This ring contains the nonzero nilpotent class $\bar{x}$, so it is not reduced.

Thus the ideal (x^2) is not radical.

Its radical is

$$\sqrt{(x^2)} = (x).$$

Example 4. Quotient that becomes a domain

Take

$$k[x, y]/(y - x^2).$$

This is isomorphic to $k[x]$, hence a domain.

Therefore $(y - x^2)$ is prime.

This is an excellent way to prove ideals are prime: show the quotient is a domain.

Example 5. Quotient that becomes reducible

Take

$$k[x, y]/(xy).$$

Here $\bar{x}\bar{y} = 0$, so the quotient has zero-divisors. Hence (xy) is not prime.

Geometrically this is the union of the two axes.

So zero-divisors often reflect reducibility.

Example 6. Center of a matrix algebra

Let

$$R = M_2(k).$$

A general matrix

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

commutes with every matrix only when $b = c = 0$ and $a = d$. Hence

$$Z(R) = \left\{ \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} : \lambda \in k \right\} = kI_2.$$

So the center isolates the true scalars inside a noncommutative algebra.

Example 7. Annihilator and associated prime

Let

$$R = k[x, y, z], \quad I = (x^2, xy, xz, yz).$$

Then in R/I ,

$$\text{Ann}_R(\bar{x}) = (x, y, z).$$

So (x, y, z) is an associated prime of R/I .

This is exactly how annihilators enter primary decomposition problems.

9. Big-picture summary

These ideas fit together very tightly:

polynomial ring $\longrightarrow$ ideal $\longrightarrow$ quotient algebra $\longrightarrow$ annihilator / support / associated primes.

And in the broader ring-theoretic setting:

$$\text{ring} \longrightarrow \text{center} \longrightarrow \text{algebra over a base field/ring}.$$

The most important guiding intuition is:

- polynomial rings are free places to write variables;
- ideals are systems of algebraic relations;
- quotients are rings where those relations hold;
- annihilators detect what acts as zero;
- centers detect what acts like scalars;
- algebras are rings built over a chosen base ring or field.

10. Compact theorem list worth remembering

$$A[x]/(x - a) \cong A$$

$$A[x]/I[x] \cong (A/I)[x]$$

$$A/\mathfrak{p} \text{ is a domain} \iff \mathfrak{p} \text{ is prime}$$

$$A/\mathfrak{m} \text{ is a field} \iff \mathfrak{m} \text{ is maximal}$$

$$\text{Ann}_A(A/I) = I$$

$$\text{Ann}_A(\bar{f}) = (I : f)$$

$$Z(M_n(k)) = kI_n$$

$k[x_1, \dots, x_n]$ is the free commutative k -algebra on $x_1, \dots, x_n$.

What are x and y in $A[x, y]$? Are they numbers?

In the polynomial ring

$$A[x, y],$$

the coefficients belong to A , but the symbols x and y do *not* initially belong to A .

They are **formal variables** (or **indeterminates**).

Thus

$$A[x, y] = \left\{ \sum_{i,j \geq 0} a_{ij} x^i y^j \mid a_{ij} \in A, \text{ only finitely many } a_{ij} \neq 0 \right\}.$$

So x and y are not yet specific numbers and not arbitrary sets. They are new formal symbols used to build a larger ring from A .

Important point:

$$x, y \in A[x, y], \quad \text{but usually} \quad x, y \notin A.$$

1. Inside the ring $A[x, y]$

Inside $A[x, y]$, the symbols x and y are abstract algebraic placeholders. They can be added and multiplied:

$$x + y, \quad x^2 y + 3y^5 + 7,$$

but one is not yet assigning them concrete values.

So the expression

$$x^2 + y$$

is not yet a number and not yet a function; it is simply an element of the ring $A[x, y]$.

2. What is evaluation?

Evaluation means substituting actual elements for the formal variables.

If $a, b \in A$, then there is an evaluation map

$$\text{ev}_{(a,b)} : A[x, y] \rightarrow A, \quad f(x, y) \mapsto f(a, b).$$

For example, if

$$f(x, y) = 2x^2 + 3xy + y + 5 \in \mathbb{Z}[x, y],$$

then

$$f(1, 2) = 2 \cdot 1^2 + 3 \cdot 1 \cdot 2 + 2 + 5 = 15.$$

So evaluation turns a formal polynomial into an actual element of the target ring.

3. Must x and y become numbers?

Not always.

In ordinary evaluation into $\mathbb{R}$, $\mathbb{C}$, or $\mathbb{Z}$, one substitutes numbers. But in general, one may evaluate in any A -algebra B .

That is, if B is an A -algebra and $b_1, b_2 \in B$, then there is a unique A -algebra homomorphism

$$A[x, y] \rightarrow B, \quad x \mapsto b_1, \quad y \mapsto b_2.$$

So the substitutions do *not* have to be numbers in the ordinary sense. They only have to be elements of a ring or algebra where addition and multiplication make sense.

4. Example: evaluation at numbers

In $\mathbb{R}[x, y]$, let

$$f(x, y) = x^2 - y.$$

Then evaluating at $(3, 4)$ gives

$$f(3, 4) = 9 - 4 = 5.$$

5. Example: evaluation at matrices

The target need not be a field of numbers.

For example, one may define

$$\varphi : \mathbb{R}[t] \rightarrow M_2(\mathbb{R}), \quad t \mapsto \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}.$$

Then

$$t^2 + 1$$

is sent to

$$\begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 5 \end{pmatrix}.$$

So the value of a polynomial need not be a number; it may be a matrix or another algebraic object.

6. Can x and y be sets?

Not in the usual meaning of polynomial evaluation.

A polynomial can only be evaluated at elements of a structure where addition and multiplication are defined. A raw set does not usually have such operations.

So if by “set” one means just an arbitrary set, then no.

If a set is equipped with suitable algebraic structure and its elements lie in a ring or algebra, then those *elements* may be substituted into the polynomial.

7. Best summary

- In $A[x, y]$, the symbols x and y are **formal indeterminates**.
- They are elements of the polynomial ring $A[x, y]$, but usually not elements of A .
- A polynomial is initially a formal expression, not yet a number.
- **Evaluation** means replacing x and y by actual elements of a target ring or algebra.
- In familiar cases these are numbers, but in general they may also be matrices, residue classes, operators, or other algebraic elements.

Ideals and quotients in polynomial rings

Let

$$R = A[x_1, \dots, x_n]$$

be a polynomial ring.

1. Ideals in a polynomial ring

An **ideal** $I \subseteq R$ is a subset such that:

1. if $f, g \in I$, then $f + g \in I$,
2. if $f \in I$ and $h \in R$, then $hf \in I$.

So an ideal is a collection of polynomials closed under:

- addition,
- subtraction,
- multiplication by arbitrary polynomials.

If

$$I = (f_1, \dots, f_r),$$

then I consists of all expressions

$$h_1 f_1 + \dots + h_r f_r, \quad h_i \in R.$$

Thus an ideal contains:

- the chosen generators $f_1, \dots, f_r$,
- all polynomial consequences of those generators.

That is why ideals represent **relations** or **systems of equations**.

2. Examples of ideals

Example 1. In $k[x]$,

$$(x) = \{xg(x) : g(x) \in k[x]\},$$

the ideal of all polynomials divisible by x .

Example 2. In $k[x, y]$,

$$(x, y) = \{xf(x, y) + yg(x, y) : f, g \in k[x, y]\}.$$

This is exactly the set of all polynomials with zero constant term.

Example 3. In $k[x, y]$,

$$(y - x^2)$$

is the ideal of all multiples of $y - x^2$. Geometrically, this corresponds to the relation

$$y = x^2.$$

3. Quotients in a polynomial ring

If $I \subseteq R$ is an ideal, then the **quotient ring**

$$R/I$$

is the ring in which every polynomial belonging to I is declared to be equal to 0.

So quotienting by I means:

$$f \in I \implies f = 0 \text{ in } R/I.$$

This is why quotients are described as **imposing relations**.

4. Main intuition

If

$$R = k[x, y], \quad I = (y - x^2),$$

then in the quotient ring

$$R/I,$$

we force

$$y - x^2 = 0,$$

so effectively

$$y = x^2.$$

Thus every polynomial in x and y can be simplified using the relation $y = x^2$. That is why

$$k[x, y]/(y - x^2) \cong k[x].$$

5. Examples of quotient rings

Example 1.

$$k[x]/(x - a).$$

Here we impose

$$x = a,$$

so the quotient is isomorphic to k .

Example 2.

$$k[x]/(x^2).$$

Here we impose

$$x^2 = 0.$$

Thus the class $\bar{x}$ is a nonzero nilpotent element.

Example 3.

$$k[x, y]/(x, y).$$

Here we impose

$$x = 0, \quad y = 0.$$

So every polynomial reduces to its constant term, and

$$k[x, y]/(x, y) \cong k.$$

Example 4.

$$k[x, y]/(xy).$$

Here we impose

$$xy = 0.$$

So in the quotient, the product of the classes of x and y is zero.

6. Geometric meaning

In

$$k[x_1, \dots, x_n],$$

an ideal I defines the algebraic set

$$V(I) = \{a \in k^n : f(a) = 0 \text{ for all } f \in I\}.$$

So:

- the ideal is the algebraic side,
- the quotient ring

$$k[x_1, \dots, x_n]/I$$

is the coordinate ring of the geometric object defined by I .

For example, if

$$I = (y - x^2),$$

then $V(I)$ is the parabola $y = x^2$, and the quotient

$$k[x, y]/(y - x^2)$$

is its coordinate ring.

7. Best short summary

- An **ideal** in a polynomial ring is a set of polynomials closed under addition and multiplication by arbitrary polynomials.
- Ideals represent polynomial relations.
- A **quotient ring** R/I is obtained by forcing all polynomials in I to become zero.
- Thus ideals are the relations, and quotients are the new algebraic worlds where those relations hold.

Why is $\bar{x} \neq 0$ in $k[x]/(x^2)$? What are the elements of the ideal and of the quotient?

Consider the quotient ring

$$k[x]/(x^2).$$

In this quotient, we force only the relation

$$x^2 = 0,$$

not the relation

$$x = 0.$$

These are different statements.

1. Why $\bar{x} \neq 0$

In the quotient, the element represented by x is the coset

$$\bar{x} = x + (x^2).$$

Now:

$$\bar{x}^2 = 0,$$

because

$$x^2 \in (x^2).$$

However,

$$\bar{x} \neq 0,$$

because $x \notin (x^2)$.

Indeed, if $\bar{x} = 0$, then x would have to belong to the ideal (x^2) . But (x^2) consists of multiples of x^2 , and x is not a multiple of x^2 in $k[x]$.

Thus

$$\bar{x} \neq 0, \quad \bar{x}^2 = 0.$$

An element with this property is called **nilpotent**.

2. What are the elements of the ideal (x^2) ?

The ideal

$$(x^2) \subseteq k[x]$$

consists of all multiples of x^2 :

$$(x^2) = \{x^2 g(x) : g(x) \in k[x]\}.$$

So its elements look like

$$0, \quad x^2, \quad 3x^2, \quad x^3, \quad x^2 + x^5, \quad 7x^2 - 4x^4, \dots$$

Every polynomial in (x^2) has zero constant term and zero x -term.

For example:

$$x^2 + 5x^3 \in (x^2), \quad x \notin (x^2), \quad 1 + x \notin (x^2).$$

3. What are the elements of the quotient $k[x]/(x^2)$?

The elements of the quotient are **cosets**

$$f(x) + (x^2).$$

This means that two polynomials are identified whenever their difference lies in (x^2) :

$$f(x) \sim g(x) \iff f(x) - g(x) \in (x^2).$$

Since all terms involving $x^2, x^3, x^4, \dots$ vanish modulo (x^2) , every coset is determined only by:

- the constant term,
- the coefficient of x .

Therefore every element of $k[x]/(x^2)$ has the form

$$a + b\bar{x}, \quad a, b \in k,$$

where

$$\bar{x}^2 = 0.$$

So, as a set,

$$k[x]/(x^2) = \{a + b\bar{x} : a, b \in k\}.$$

4. Examples of equality in the quotient

In $k[x]/(x^2)$,

$$x^2 + x + 3 = x + 3,$$

because their difference is

$$x^2 \in (x^2).$$

Also,

$$5x^4 + 2x^2 + x + 7 = x + 7,$$

because

$$5x^4 + 2x^2 \in (x^2).$$

So all powers $x^2, x^3, x^4, \dots$ disappear in the quotient.

5. How addition and multiplication work

If elements are written as

$$a + b\bar{x},$$

then addition is

$$(a + b\bar{x}) + (c + d\bar{x}) = (a + c) + (b + d)\bar{x}.$$

Multiplication is

$$(a + b\bar{x})(c + d\bar{x}) = ac + (ad + bc)\bar{x} + bd\bar{x}^2.$$

Since $\bar{x}^2 = 0$, this simplifies to

$$(a + b\bar{x})(c + d\bar{x}) = ac + (ad + bc)\bar{x}.$$

6. Comparison with $k[x]/(x)$

In

$$k[x]/(x),$$

we impose

$$x = 0,$$

so every element reduces to a constant and

$$k[x]/(x) \cong k.$$

But in

$$k[x]/(x^2),$$

we impose only

$$x^2 = 0,$$

so the class $\bar{x}$ survives as a nonzero nilpotent element.

Hence $k[x]/(x^2)$ is larger than k .

7. Best intuition

The quotient

$$k[x]/(x^2)$$

is the ring where x behaves like an *infinitesimal*:

$$\bar{x} \neq 0, \quad \bar{x}^2 = 0.$$

So it is not the same as setting $x = 0$. Rather, one keeps a first-order term in x , while all higher powers vanish.

8. Short summary

$$(x^2) = \{x^2 g(x) : g(x) \in k[x]\}$$

is the ideal of all multiples of x^2 .

The quotient

$$k[x]/(x^2)$$

consists of cosets $f(x) + (x^2)$, and every element can be written uniquely as

$$a + b\bar{x}.$$

In this quotient,

$$\bar{x} \neq 0, \quad \bar{x}^2 = 0.$$